

FastNBessel: Efficient transforms with arbitrary products of Bessel functions and their cosmological applications

Xiao Fang^{1,*}

¹*Prospect Park Research, New York, NY 11215, USA*

(Dated: October 3, 2026)

Products of Bessel functions appear in correlation functions, higher-order statistics, and covariances in cosmology, but their oscillatory integrals are costly to evaluate directly. We present FastNBessel, an FFTLog-based method for an arbitrary number N_B of spherical Bessel functions and d integration variables. A d -dimensional FFTLog expands the input in complex power laws, after which each integral separates into multiple-Bessel Mellin transforms. We calculate each transform as a linear convolution of single-Bessel gamma-function kernels using FFTs, without evaluating hypergeometric functions. Derivatives, bin averages with fixed logarithmic widths, and cylindrical Bessel functions follow from modified kernels. For equal grids of size N and fixed N_B , the cost on a full output grid is $\mathcal{O}(dN^d \log N)$, compared with $\mathcal{O}(N^{2d})$ for brute-force quadrature and $\mathcal{O}(dN^{d+1})$ for an optimized quadrature using tensor contraction. We demonstrate the method for integrals in galaxy and shear three-point functions, covariances, the spherical Fourier–Bessel bispectrum, and two-loop perturbation theory, and validate it against direct quadrature. The FastNBessel code package is released along with this paper.

I. INTRODUCTION

Current and upcoming cosmological surveys, including DESI [1], Euclid [2], the Vera C. Rubin Observatory Legacy Survey of Space and Time [3], SPHEREx [4], the Nancy Grace Roman Space Telescope [5], and CMB-S4 [6], will measure the large-scale structure over large volumes and wide redshift ranges. The statistical uncertainties of these measurements will be sufficiently small that the theoretical predictions must be both accurate and efficient enough to be used in parameter inference.

The two-point function contains all the statistical information of a Gaussian field. However, nonlinear gravitational evolution and galaxy bias render the late-time galaxy density field non-Gaussian, while redshift-space distortions further complicate its statistics. Higher-order statistics, such as the bispectrum and three-point correlation function, contain information that is not present in the two-point function. Connected four-point and higher-order functions also contribute to the covariances of lower-order statistics. These statistics can be used to study galaxy bias, primordial non-Gaussianity, gravity, and the growth of structure; see Ref. [7] for a review. Joint full-shape analyses of the power spectrum and bispectrum have been performed with BOSS [8] and, more recently, DESI [9, 10], providing examples of the additional constraining power. Modeling these observables on smaller scales requires calculations beyond linear theory and accurate non-Gaussian covariances.

Bessel integrals commonly appear in these calculations. Under spherical symmetry, the plane-wave expansion separates the angular and radial dependences, and the radial transforms contain spherical Bessel functions $j_\ell(kr)$. Products of spherical Bessel functions occur in multi-

point correlation functions, two-loop calculations, projected polyspectra, and covariances. For quantities with cylindrical symmetry, such as flat-sky angular correlation functions, the corresponding Hankel transforms contain cylindrical Bessel functions $J_n(\ell\theta)$. Depending on the application, several Bessel functions may share one integration variable, or different groups of Bessel functions may depend on different integration variables.

A direct quadrature of these integrals is often slow to converge because the Bessel functions oscillate rapidly. The oscillation frequency increases with the Bessel argument, and the final result may be much smaller than the positive and negative contributions to the integral. The calculation becomes particularly expensive when the transform is needed at many separations or on a multidimensional output grid. Adaptive quadrature and methods designed for oscillatory integrals, such as Levin integration, are useful for individual points [11, 12], but many such integrations are required for a dense output grid.

The FFTLog algorithm evaluates logarithmically sampled Hankel transforms by decomposing the smooth part of the integrand into complex power laws [13, 14]. It has been applied to nonlinear perturbation theory, angular power spectra and bispectra, and projected correlation functions [15–26]. The FFTLog kernel was also extended to derivatives of spherical Bessel functions in Ref. [24]. For an integral with two Bessel functions, the integral of each power-law component is often written in terms of a Gauss hypergeometric function. Numerical evaluation of this function requires special treatment near equal arguments and in different analytic regions. The 2D-FFTLog method avoids this problem when the two Bessel functions depend on two independent integration variables, and has been used to calculate real-space covariance matrices [27]. Products of spherical Bessel functions have also been treated by trigonometric decompositions [28], analytic recursions and distributional formulae [29, 30], and specialized quadrature for up to three Bessel functions

* xfangcosmo@gmail.com

[12].

In this paper, we present a new FFTLog method for integrals containing an arbitrary number of Bessel functions. We use the Mellin product theorem to write the Mellin transform of a product of Bessel functions as a convolution of the single-Bessel transforms. The convolution is calculated with FFTs, so only gamma-function ratios are needed and no hypergeometric function is evaluated. We then generalize this result to d integration variables by using a d -dimensional FFTLog for the input function. The input function need not be separable, as long as each Bessel function depends on one of the integration variables.

This paper is organized as follows. In Sec. II, we derive the one-dimensional and multidimensional algorithms and discuss Bessel derivatives, bin averaging, and other types of Bessel functions. In Sec. III, we discuss applications to three-point functions, covariance matrices, the spherical Fourier-Bessel bispectrum, and two-loop power spectrum calculations. In Sec. IV, we discuss the numerical parameters and computational scaling, and summarize our results. Useful Mellin-transform identities and thorough numerical tests are given in the appendices.

II. METHOD

A. One-dimensional integrals with a common x

We first consider an integral with $N_B = p$ spherical Bessel functions that share the integration variable x ,

$$I(y; \mathbf{t}) = \int_0^\infty \frac{dx}{x} f(x) \prod_{a=1}^p j_{\ell_a}(t_a x y), \quad t_1 \equiv 1. \quad (1)$$

For a fixed $\mathbf{t} = (1, t_2, \dots, t_p)$, the algorithm calculates $I(y; \mathbf{t})$ at all points on a logarithmic y grid. These points form the ray $(y_1, \dots, y_p) = y(1, t_2, \dots, t_p)$ in the space of Bessel arguments.

Let

$$x_q = x_0 e^{q\Delta}, \quad y_n = y_0 e^{n\Delta}, \quad \eta_s = \frac{2\pi s}{N\Delta}, \quad (2)$$

where $q, n = 0, \dots, N-1$ and Δ is the common logarithmic spacing, $\Delta = \ln(x_{q+1}/x_q) = \ln(y_{n+1}/y_n)$. We take N to be even and label the N Fourier modes by $s = -N/2, \dots, N/2 - 1$.¹ For an FFTLog bias ν , we

write

$$\tilde{c}_s = \sum_{q=0}^{N-1} \frac{f(x_q)}{x_q^\nu} e^{-2\pi i s q/N}, \quad c_s = W_s \tilde{c}_s, \quad (3)$$

$$f(x) \simeq \frac{1}{N} \sum_{s=-N/2}^{N/2-1} c_s x_0^{-i\eta_s} x^{\nu+i\eta_s}. \quad (4)$$

Here W_s is an optional window function used to reduce ringing caused by the finite logarithmic interval. Without the window, Eq. (4) exactly reconstructs the sampled values at $x = x_q$ and gives the usual periodic FFTLog interpolation between them. In our implementation the Nyquist coefficient is set to zero, $c_{-N/2} = 0$.

Substituting Eq. (4) into Eq. (1), the integral of each power-law component is the Mellin transform

$$\mathcal{M}_p(z; \mathbf{t}) = \int_0^\infty \frac{du}{u} u^z \prod_{a=1}^p j_{\ell_a}(t_a u). \quad (5)$$

Useful Mellin-transform identities are collected in Appendix A.

The single-Bessel Mellin transform is

$$U_\ell(z) \equiv \int_0^\infty \frac{du}{u} u^z j_\ell(u) = \frac{\sqrt{\pi}}{4} g_\ell(z), \quad (6)$$

$$g_\ell(z) = 2^z \frac{\Gamma[(\ell+z)/2]}{\Gamma[(3+\ell-z)/2]}, \quad -\ell < \text{Re } z < 2. \quad (7)$$

We divide the total bias among the p Bessel functions,

$$\nu = \sum_{a=1}^p \beta_a, \quad -\ell_a < \beta_a < 2, \quad (8)$$

where each β_a is chosen inside the fundamental strip of the single-Bessel transform. We now derive the Mellin transform of the product. Define

$$h_a(u) = j_{\ell_a}(t_a u), \quad H_a(z) = \int_0^\infty \frac{du}{u} u^z h_a(u) = t_a^{-z} U_{\ell_a}(z), \quad (9)$$

The inverse Mellin transform is

$$h_a(u) = \frac{1}{2\pi i} \int_{\beta_a - i\infty}^{\beta_a + i\infty} dz_a u^{-z_a} H_a(z_a). \quad (10)$$

We insert Eq. (10) for the first $p-1$ Bessel functions into Eq. (5). Each z_a contour is along $\text{Re } z_a = \beta_a$. After exchanging the order of integration, we obtain

$$\begin{aligned} \mathcal{M}_p(z; \mathbf{t}) &= \frac{1}{(2\pi i)^{p-1}} \int dz_1 \cdots \int dz_{p-1} \\ &\quad \times \prod_{a=1}^{p-1} H_a(z_a) \int_0^\infty \frac{du}{u} u^{z - \sum_{a=1}^{p-1} z_a} h_p(u) \\ &= \frac{1}{(2\pi i)^{p-1}} \int dz_1 \cdots \int dz_{p-1} \\ &\quad \times \prod_{a=1}^{p-1} H_a(z_a) H_p\left(z - \sum_{a=1}^{p-1} z_a\right). \end{aligned} \quad (11)$$

¹ In our previous papers [16, 18, 24, 27] we used $s = -N/2, \dots, N/2$, with the two representations of the Nyquist mode identified or half weighted. The two conventions are equivalent; the range used here simply keeps one copy of the Nyquist mode.

Thus the Mellin transform of the product is a convolution of the individual Mellin transforms.

We next set $z = \nu + i\eta_s$ and write $z_a = \beta_a + i\eta_a$ for $a < p$, and define $\beta_p = \nu - \sum_{a < p} \beta_a$. The argument of the last transform is then

$$z_p = z - \sum_{a < p} z_a = \beta_p + i \left(\eta_s - \sum_{a < p} \eta_a \right). \quad (12)$$

We define the kernel on each vertical contour as

$$R_a(\eta) = t_a^{-\beta_a - i\eta} U_{\ell_a}(\beta_a + i\eta). \quad (13)$$

Using $dz_a = i d\eta_a$, Eq. (11) becomes

$$\begin{aligned} \mathcal{M}_p(\nu + i\eta_s; \mathbf{t}) &= \frac{1}{(2\pi)^{p-1}} \int_{-\infty}^{\infty} d\eta_1 \cdots \int_{-\infty}^{\infty} d\eta_{p-1} \\ &\times \prod_{a=1}^{p-1} R_a(\eta_a) R_p \left(\eta_s - \sum_{a < p} \eta_a \right). \end{aligned} \quad (14)$$

We sample the contours at $\eta_m = \lambda m$, where $\lambda = 2\pi/(N\Delta)$ in the basic implementation. Applying the trapezoidal rule at the outer mode $\eta_s = \lambda s$ gives

$$\begin{aligned} \mathcal{M}_p(\nu + i\eta_s; \mathbf{t}) &\simeq \left(\frac{\lambda}{2\pi} \right)^{p-1} \sum_{m_1, \dots, m_{p-1} \in \mathbb{Z}} \prod_{a=1}^{p-1} R_a(\lambda m_a) \\ &\times R_p \left[\lambda \left(s - \sum_{a < p} m_a \right) \right]. \end{aligned} \quad (15)$$

We define $m_p = s - \sum_{a < p} m_a$, so that $m_1 + \cdots + m_p = s$. Using $U_\ell = (\sqrt{\pi}/4)g_\ell$, we further define

$$Q_a(m) = t_a^{-\beta_a - i\lambda m} g_{\ell_a}(\beta_a + i\lambda m). \quad (16)$$

Equation (15) can then be written as

$$\boxed{\mathcal{M}_p(\nu + i\eta_s; \mathbf{t}) \simeq \left(\frac{\lambda}{2\pi} \right)^{p-1} \left(\frac{\sqrt{\pi}}{4} \right)^p [Q_1 * \cdots * Q_p](s)}. \quad (17)$$

Here the linear discrete convolution is defined by

$$[Q_1 * \cdots * Q_p](s) \equiv \sum_{m_1 + \cdots + m_p = s} \prod_{a=1}^p Q_a(m_a). \quad (18)$$

Equation (17) uses the same frequency spacing for the Mellin convolution and the outer FFTLog. We may instead oversample the Mellin integral by an integer factor r_η . In this case the internal spacing is $\lambda_{\text{int}} = \lambda/r_\eta$, and the outer mode s is obtained from the convolution mode $r_\eta s$:

$$\mathcal{M}_p(\nu + i\eta_s; \mathbf{t}) \simeq \left(\frac{\lambda_{\text{int}}}{2\pi} \right)^{p-1} \left(\frac{\sqrt{\pi}}{4} \right)^p [Q_1 * \cdots * Q_p](r_\eta s). \quad (19)$$

For a truncated internal lattice $m = -M, \dots, M$ of length $N_\eta = 2M + 1$, the full p -fold linear convolution has length

$$L_{\text{conv}} = p(N_\eta - 1) + 1. \quad (20)$$

We evaluate this linear convolution by zero-padding each Q_a array to a length $L \geq L_{\text{conv}}$, Fourier transforming the arrays, multiplying them, and applying an inverse FFT. Without the zero-padding, a circular convolution would alias the high-frequency tails into the modes of interest.

Finally, substituting Eq. (4) into Eq. (1) and changing variables to $u = xy$ gives

$$I(y; \mathbf{t}) \simeq \frac{y^{-\nu}}{N} \sum_{s=-N/2}^{N/2-1} c_s(x_0 y)^{-i\eta_s} \mathcal{M}_p(\nu + i\eta_s; \mathbf{t}). \quad (21)$$

At $y_n = y_0 e^{n\Delta}$, the y -dependent phase is $(y_n/y_0)^{-i\eta_s} = e^{-2\pi i s n/N}$. We reverse the spectral index, or use Hermitian symmetry when the input is real, and calculate the sum with one inverse FFT:

$$\boxed{I(y_n; \mathbf{t}) \simeq y_n^{-\nu} \text{IFFT} [c_s^*(x_0 y_0)^{i\eta_s} \mathcal{M}_p^*(\nu + i\eta_s; \mathbf{t})]_n} \quad (22)$$

The calculation therefore consists of an FFTLog of the input [Eq. (3)], the Mellin convolution [Eq. (17)], and an inverse FFTLog of the output [Eq. (22)]. The Fourier convention and normalization are given in Appendix B.

For $p = 1$, Eq. (17) contains no internal convolution and reduces exactly to the usual one-Bessel FFTLog kernel in Eq. (6). For $p = 2$, it becomes a single convolution of two gamma-function ratios and evaluates the two-Bessel transform along $(y_1, y_2) = y(1, t_2)$. Unlike the standard analytic result for a power-law input, this calculation does not require a Gauss hypergeometric function.

B. The general multidimensional transform

We now generalize the calculation to d integration variables. Let p_α Bessel functions depend on x_α ($p_\alpha \geq 1$), so that

$$N_B = \sum_{\alpha=1}^d p_\alpha. \quad (23)$$

We consider the transform

$$\mathcal{T}(\mathbf{y}; \mathbf{t}) = \int_0^\infty \Phi(\mathbf{x}) \prod_{\alpha=1}^d \left[\frac{dx_\alpha}{x_\alpha} \prod_{a=1}^{p_\alpha} j_{\ell_{\alpha a}}(t_{\alpha a} x_\alpha y_\alpha) \right], \quad (24)$$

where $t_{\alpha 1} = 1$. The function Φ does not need to be separable in the x_α . Its dependence on all d variables is represented by a d -dimensional FFTLog.

For logarithmic grids $x_{\alpha q} = x_{\alpha 0} \exp(q\Delta_\alpha)$ and $y_{\alpha n} = y_{\alpha 0} \exp(n\Delta_\alpha)$, define

$$\eta_{\alpha s} = \frac{2\pi s}{N_\alpha \Delta_\alpha}, \quad N_{\text{grid}} = \prod_{\alpha=1}^d N_\alpha. \quad (25)$$

With outer biases $\boldsymbol{\nu} = (\nu_1, \dots, \nu_d)$, the multidimensional coefficients are

$$\tilde{c}_{\mathbf{s}} = \sum_{q_1 \dots q_d} \frac{\Phi(x_{1q_1}, \dots, x_{dq_d})}{\prod_{\alpha} x_{\alpha q_{\alpha}}^{\nu_{\alpha}}} \exp \left[-2\pi i \sum_{\alpha} \frac{s_{\alpha} q_{\alpha}}{N_{\alpha}} \right], \quad (26)$$

with an optional window $c_{\mathbf{s}} = W_{\mathbf{s}} \tilde{c}_{\mathbf{s}}$. The corresponding power-law expansion is

$$\begin{aligned} \Phi(\mathbf{x}) &\simeq \frac{1}{N_{\text{grid}}} \sum_{s_1=-N_1/2}^{N_1/2-1} \dots \sum_{s_d=-N_d/2}^{N_d/2-1} c_{\mathbf{s}} \\ &\times \prod_{\alpha=1}^d x_{\alpha 0}^{-i\eta_{\alpha s_{\alpha}}} x_{\alpha}^{\nu_{\alpha} + i\eta_{\alpha s_{\alpha}}}. \end{aligned} \quad (27)$$

After this expansion is substituted into Eq. (24), the d integrations separate for each Fourier mode. For the Bessel functions that depend on x_{α} , we calculate

$$K_{\alpha s} = \mathcal{M}_{p_{\alpha}}(\nu_{\alpha} + i\eta_{\alpha s}; \mathbf{t}_{\alpha}) \quad (28)$$

using Eq. (17), with $\nu_{\alpha} = \sum_a \beta_{\alpha a}$. We absorb the grid phase and complex conjugation into $\hat{K}_{\alpha s} \equiv (x_{\alpha 0} y_{\alpha 0})^{i\eta_{\alpha s}} K_{\alpha s}^*$. The full kernel is the outer product $\prod_{\alpha} \hat{K}_{\alpha s_{\alpha}}$, while the coefficient $c_{\mathbf{s}}$ contains the nonseparable dependence of the input. The output is then obtained with one inverse d -dimensional FFT,

$$\boxed{\mathcal{T}_{\mathbf{n}} \simeq \prod_{\alpha} y_{\alpha n_{\alpha}}^{-\nu_{\alpha}} \text{IFFT}_d \left[c_{\mathbf{s}}^* \prod_{\alpha} \hat{K}_{\alpha s_{\alpha}} \right]_{\mathbf{n}}}. \quad (29)$$

The Mellin convolutions remain one dimensional. They can be calculated once for each group and reused for different input functions on the same grids.

For $d = 2$ and $(p_1, p_2) = (1, 1)$, each group kernel is the ordinary single-Bessel Mellin transform. Equations (26) and (29) then reduce to the 2D-FFTLog algorithm of Ref. [27]. The one-dimensional method in Sec. II A is recovered by setting $d = 1$ and $p_1 = N_B$. Convergence conditions for the Mellin contours are discussed in Appendix C.

C. Derivatives and bin averaging

A derivative of a Bessel function can be included without changing the algorithm. Repeated integration by parts gives

$$U_{\ell}^{[q]}(z) = \int_0^{\infty} \frac{du}{u} u^z j_{\ell}^{(q)}(u) = (-1)^q \frac{\Gamma(z)}{\Gamma(z-q)} U_{\ell}(z-q), \quad (30)$$

so for a q th derivative we replace $g_{\ell}(z)$ in Eq. (16) by $g_{\ell}^{[q]}(z) = (-1)^q \Gamma(z) g_{\ell}(z-q) / \Gamma(z-q)$.

We can also average the Bessel function over a bin whose edges are fixed multiples of the output scale. In

dimension D , define

$$\bar{j}_{\ell}(u; a, b) = \frac{D}{b^D - a^D} \int_a^b dv v^{D-1} j_{\ell}(vu). \quad (31)$$

Its Mellin transform is $B_D(z; a, b) U_{\ell}(z)$, where

$$B_D(z; a, b) = \frac{D}{b^D - a^D} \frac{b^{D-z} - a^{D-z}}{D-z}. \quad (32)$$

The apparent singularity at $z = D$ is removable and we use its analytic limit. Thus derivatives and bins of fixed logarithmic width only modify the corresponding $Q_{\alpha a}$ kernel. Derivations of both results are given in Appendix D.

D. Cylindrical Bessel functions

For flat-sky angular statistics, the spherical Bessel function is replaced by the cylindrical Bessel function J_n . We can use $J_n(u) = \sqrt{2u/\pi} j_{n-1/2}(u)$ and absorb the additional power of u into the input and the FFTLog bias, as in Ref. [27]. Alternatively, we can use the Mellin transform

$$V_n(z) = \int_0^{\infty} \frac{du}{u} u^z J_n(u) = 2^{z-1} \frac{\Gamma[(n+z)/2]}{\Gamma[(2+n-z)/2]}. \quad (33)$$

The rest of the calculation is unchanged after U_{ℓ} is replaced by V_n . Bin averaging also gives the same factor $B_D(z; a, b)$ in Eq. (32); for an area-weighted angular bin, $D = 2$. For integer orders, negative values can first be replaced using $J_{-n}(u) = (-1)^n J_n(u)$; see Appendix D.

E. Other Bessel families

The same method applies whenever the Mellin transform of each individual function can be evaluated stably. Define

$$U_{\ell}^{(A)}(z) = \int_0^{\infty} \frac{du}{u} u^z \chi_{\ell}^{(A)}(u). \quad (34)$$

For the spherical Neumann function y_{ℓ} and the modified spherical Bessel function $k_{\ell}(u) = \sqrt{\pi/(2u)} K_{\ell+1/2}(u)$, standard Bessel integrals give the following expressions [31, 32]:

$$U_{\ell}^{(y)}(z) = \frac{\sqrt{\pi}}{4} (-1)^{\ell+1} 2^z \frac{\Gamma[(z-\ell-1)/2]}{\Gamma[(2-\ell-z)/2]}, \quad (35)$$

$$U_{\ell}^{(k)}(z) = \frac{\sqrt{\pi}}{4} 2^{z-1} \Gamma\left(\frac{z-\ell-1}{2}\right) \Gamma\left(\frac{z+\ell}{2}\right). \quad (36)$$

The fundamental strips are $\ell+1 < \Re z < 2$ for y_{ℓ} and $\Re z > \ell+1$ for k_{ℓ} . The former is empty for $\ell \geq 1$, so y_{ℓ} requires a lower cutoff of the integration range, subtraction of its singular small- u terms, or analytic continuation. The spherical Hankel functions $h_{\ell}^{(1,2)} = j_{\ell} \pm i y_{\ell}$ are linear

combinations of the two Mellin kernels and inherit the same small- u issue.

The modified spherical Bessel function $i_\ell(u)$ grows as $e^u/(2u)$ and has no convergent parameter space for the infinite Mellin transform integral. Once the transform of each individual function is given, the convolution algorithm is unchanged.

Products of modified Bessel K functions occur in momentum-space CFT and in correlators constrained by conformal symmetry, such as triple- K integrals [33, 34], which have applications in some inflationary models; see, e.g., Ref. [35]. We leave the necessary regularization, implementation, and numerical tests for these additional families to future work.

III. COSMOLOGICAL APPLICATIONS

Multiple-Bessel transforms enter a number of cosmological calculations. We give four classes of examples below and identify the part of each calculation that can be evaluated with FastNBessel. Table I summarizes the transforms. The references cited below are representative examples rather than a comprehensive review of each literature.

For the numerical examples in this section, we use a flat Λ CDM model with $\Omega_m = 0.315$, $\Omega_b = 0.049$, $h = 0.674$, $n_s = 0.965$, and $\sigma_8 = 0.811$ (Planck 2018 Λ CDM TT+TE+EE+low- ℓ pol+CMB lensing [36]). We use the Eisenstein–Hu no-wiggle linear power spectrum $P_L(k, z)$ [37] and the tree-level matter bispectrum [7]

$$B_m(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3; z) = 2F_2(\mathbf{k}_1, \mathbf{k}_2)P_L(k_1, z)P_L(k_2, z) + 2 \text{ cyclic permutations}, \quad (37)$$

with the usual second-order kernel F_2 , and a local Eulerian galaxy-bias model [38]. For the real-space galaxy examples this gives

$$B_g = b_1^3 B_m + b_1^2 b_2 [P_L(k_1)P_L(k_2) + 2 \text{ cyclic}], \quad (38)$$

and we choose $b_1 = 2$ and $b_2 = 0.3$. Unless stated otherwise, the sources are evaluated at $z = 0.7$. We evaluate the physical model well beyond the scales contributing to each displayed result. For $0 < a < b$, define the logarithmic raised-cosine ramp

$$\mathcal{R}(x; a, b) = \begin{cases} 0, & x \leq a, \\ \frac{1}{2} \left[1 - \cos \left(\pi \frac{\ln(x/a)}{\ln(b/a)} \right) \right], & a < x < b, \\ 1, & x \geq b. \end{cases} \quad (39)$$

The endpoint window used along each input axis is

$$W_{\text{end}}(x) = \mathcal{R}(x; x_{\min}, x_{\text{lo}}) \mathcal{R}(x^{-1}; x_{\max}^{-1}, x_{\text{hi}}^{-1}), \quad (40)$$

where $x_{\min} < x_{\text{lo}} < x_{\text{hi}} < x_{\max}$. It equals unity on $[x_{\text{lo}}, x_{\text{hi}}]$ and vanishes outside $[x_{\min}, x_{\max}]$. This distant

endpoint window and explicit zero padding suppress wrap-around from the finite logarithmic grid. We enlarge the model interval and the zero-padded extension until repeating the calculation with a wider input range changes each plotted value by less than 10^{-5} of the maximum absolute value on the plotted curve. The power spectra, bispectrum multipoles, covariance source, and radial survey window functions are calculated first and saved as densely sampled text files, which are then read by FastNBessel. These simple inputs are intended to test the transforms rather than to describe a particular survey. Thorough tests with analytic input functions and timing measurements are given in Appendix E.

For all comparisons, we define the fractional difference between the FastNBessel and quadrature results as

$$\epsilon_{\text{frac}} = \frac{I_{\text{fast}} - I_{\text{quad}}}{I_{\text{quad}}}. \quad (41)$$

The numerical choices for each plotted application are collected in Table II. For the one-dimensional quadrature references, we use 131073 samples for the power-spectrum kernels and 32769 samples for the SFB kernels. The two-dimensional references use tensor-Simpson integration, with logarithmic sampling at small wavenumber and linear sampling fine enough to resolve the largest Bessel argument. Increasing either resolution satisfies the $\epsilon_Q < 10^{-5}$ convergence criterion in Eq. (E10). We use fourfold Mellin oversampling whenever two or more Bessel functions share an integration variable.

A. Galaxy and weak-lensing three-point functions

The galaxy 3PCF is the configuration-space counterpart of the bispectrum and measures the excess probability of finding galaxies in triangular configurations. Its dependence on triangle shape and orientation provides information on nonlinear growth, galaxy bias, redshift-space distortions, and primordial non-Gaussianity beyond the two-point function. For example, the large-scale 3PCF and its baryon acoustic feature have been measured in BOSS [39]. An FFT-based tripolar spherical-harmonic decomposition has also been used to measure anisotropic bispectrum multipoles from BOSS DR12 [40]. Anisotropic 3PCF multipoles have been combined with the two-point function to constrain modified gravity [41]. Full-shape two- and three-point modelling has also been developed for the Euclid spectroscopic analysis [42]. After expanding the angular dependence of the bispectrum, a redshift-space 3PCF multipole takes the form given in Eq. (35) of Ref. [43],

$$\zeta_{g, L\ell_{12}}(r_{13}, r_{23}) = i^{2\ell_{12}+L} \int_0^\infty \int_0^\infty \frac{dk_1 dk_2}{k_1 k_2} \Delta_{B, L\ell_{12}}^{\text{RSD}}(k_1, k_2) \times j_{\ell_{12}+L}(k_1 r_{13}) j_{\ell_{12}}(k_2 r_{23}), \quad (42)$$

where r_{13} and r_{23} are the lengths of two sides of the triangle formed by the galaxy triplet, and k_1 and k_2 are

TABLE I. Examples of cosmological calculations that contain the transforms considered in this paper.

Application	N_B	d	Bessel family	Additional feature
Galaxy and shear 3PCFs	2	2	j_ℓ or J_n	angular-bin averages
Anisotropic 3PCF covariance	4	2	j_ℓ	two Bessel functions per variable
Common component of galaxy N -point covariance	3	1	j_ℓ	radial-bin averages
Spherical Fourier–Bessel bispectrum	3	1	j_ℓ, j_ℓ''	derivatives
Two-loop perturbation theory	3	1	j_ℓ	–

TABLE II. FastNBessel numerical parameters for the cosmological examples. The FFT input and output intervals are the full logarithmic grids, not only the displayed ranges. Before zero padding, the galaxy-3PCF source is sampled at 1544 points over $10^{-5} \leq k \leq 50.25 h \text{ Mpc}^{-1}$ and the 3PCF-covariance source at 773 points over $10^{-5} \leq k \leq 50.76 h \text{ Mpc}^{-1}$. Adding respectively 384 and 192 zero samples at each end gives the FFT input intervals and N_x values listed below. No explicit zero samples are added in the other examples, so their physical-source and FFT input intervals coincide. N_x counts all input samples per axis, including any added zeros, and N_y counts all output samples per axis. The angular output θ is in radians. A semicolon in the β column separates integration variables. When one Bessel function depends on a variable, $\beta_{\alpha 1} = \nu_\alpha$ and its analytic kernel is evaluated directly, so r_η and N_η do not enter.

Example	FFT input interval	output interval	$N_x = N_y$	$\{\nu_\alpha\}$	$\{\beta_{\alpha\alpha}\}$	(r_η, N_η)
Galaxy 3PCF (Sec. III A)	$k_{1,2} : 2.1 \times 10^{-7} - 2.3 \times 10^3$	$r_{1,2} : 0.2 - 2.2 \times 10^9$	2312	(1, 1)	(1); (1)	–
Shear 3PCF (Sec. III A)	$\ell_{1,2} : 0.5 - 10^6$	$\theta_{1,2} : 10^{-7} - 0.2$	513	(0, 0)	(0); (0)	–
3PCF covariance (Sec. III B)	$k_{1,2} : 2.1 \times 10^{-7} - 2.4 \times 10^3$	$r_{1,2} : 0.2 - 2.2 \times 10^9$	1157	(1, 1)	(0.5, 0.5); (0.5, 0.5)	(4, 65537)
N -point covariance (Sec. III B)	$k : 10^{-7} - 10^3$	$r : 0.2 - 2 \times 10^9$	2048	1.2	(0.4, 0.4, 0.4)	(4, 65537)
SFB bispectrum (Sec. III C)	$r : 0.1 - 10^4$	$k : 2 \times 10^{-4} - 20$	8193	2	(2/3, 2/3, 2/3)	(4, 65537)
SFB window (Sec. III C)	$r : 0.1 - 10^4$	$k : 2 \times 10^{-4} - 20$	8193	1	(0.5, 0.5)	(4, 65537)
Two-loop radial block (Sec. III D)	$q : 10^{-7} - 10^3$	$r : 0.2 - 2 \times 10^9$	2048	1.2	(0.4, 0.4, 0.4)	(4, 65537)

their conjugate wavenumber magnitudes. The integer L labels the multipole with respect to the line of sight, while ℓ_{12} labels the multipole of the angle between two triangle sides. The function $\Delta_{B,L\ell_{12}}^{\text{RSD}}$ is the corresponding dimensionless redshift-space galaxy-bispectrum multipole; the subscript B denotes the bispectrum and the subscript g on ζ denotes the galaxy field. In the convention used here, $\Delta_{B,L\ell_{12}}^{\text{RSD}} = k_1^3 k_2^3 B_{g,L\ell_{12}}^{\phi_n} / (2\pi^2)^2$, where $B_{g,L\ell_{12}}^{\phi_n}$ is the galaxy-bispectrum coefficient in the tripolar spherical-harmonic basis of Ref. [43]. Equations (100)–(111) of Ref. [43] give the redshift-space galaxy bispectrum and its angular projection, while its Eq. (35) gives the normalization above and the 3PCF transform. This is the $(N_B, d) = (2, 2)$ case of Eq. (24). The bispectrum multipoles need not be separable in k_1 and k_2 , which is the reason for using the two-dimensional FFTLog. This two-Bessel transform was already evaluated with 2D-FFTLog in Ref. [43]. The same approach has also been used to obtain the next-to-leading-order matter 3PCF [44]. Related double-Bessel transforms enter bispectrum-window convolution in both Hankel-transform and tripolar spherical-harmonic formulations [45, 46].

For the galaxy example in the left panel of Fig. 1, we form the real-space bispectrum monopole

$$B_g^{(0)}(k_1, k_2) = \frac{1}{2} \int_{-1}^1 d\mu B_g(k_1, k_2, k_3(\mu)),$$

$$k_3(\mu) = \sqrt{k_1^2 + k_2^2 + 2k_1 k_2 \mu}, \quad (43)$$

where $\mu = \hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}}_2$ is the cosine of the angle between $\mathbf{k}_1$ and $\mathbf{k}_2$. We evaluate this average with 48-point Gauss–Legendre quadrature. We take $L = \ell_{12} = 0$, show $r_{13} = r_{23}$ over $20\text{--}150 h^{-1} \text{ Mpc}$, and average each separation over the volume-weighted bin $[0.95r, 1.05r]$. The physical bispectrum is evaluated over $10^{-5} < k < 50 h \text{ Mpc}^{-1}$. Raised-cosine windows are confined to $10^{-5} < k < 10^{-4} h \text{ Mpc}^{-1}$ and $20 < k < 50 h \text{ Mpc}^{-1}$, after which the input is zero-padded. Moving either taper farther from the contributing range satisfies the endpoint-convergence test described above. This real-space example does not include redshift-space distortions or shot noise.

Weak gravitational lensing measures coherent distortions of galaxy shapes by the projected matter distribution. Its two-point function probes the projected matter power spectrum, whereas the shear 3PCF is sourced by the projected matter bispectrum and is sensitive to non-linear and non-Gaussian structure. Second- and third-order shear statistics have been combined to constrain cosmology, for example with CFHTLenS [47] and DES Year 3 [48, 49]. In the latter case, the four natural shear 3PCFs and aperture-mass moments were measured in Ref. [48] and combined with the two-point functions in Ref. [49]. A shear 3PCF multipole is given in Eq. (38) of

Ref. [50],

$$\Gamma_0^{\times, (M)}(\theta_1, \theta_2) = \frac{1}{(2\pi)^3} \int_0^\infty \int_0^\infty \ell_1 \ell_2 d\ell_1 d\ell_2 \times H_M(\ell_1, \ell_2) J_{M-3}(\ell_1 \theta_1) J_{-M-3}(\ell_2 \theta_2), \quad (44)$$

where θ_1 and θ_2 are two angular side lengths of the shear triangle, and ℓ_1 and ℓ_2 are the magnitudes of the two-dimensional angular wavevectors. The integer M labels the Fourier multipole of the angle between ℓ_1 and ℓ_2 . The quantity $\Gamma_0^{\times, (M)}$ is the M th multipole of the natural shear component Γ_0 . The superscript $\times$ labels the $\times$ -projection convention of Ref. [50], in which the shear at each triangle vertex is projected relative to directions fixed by the triangle sides. Its Eqs. (41)–(43) give the other natural components, which differ in their Bessel orders and angular phases. The source H_M is obtained by expanding the convergence bispectrum in Legendre multipoles and coupling them to the Fourier multipole M with a purely geometrical function. Equations (28), (29), (35), and (39) of that reference give these steps explicitly. The geometrical coupling can be tabulated, and the multipole sum is truncated once the bispectrum multipoles become negligible. Equation (44) can therefore be calculated with the J_n kernel in Sec. II D. Area-weighted angular bins are included with $D = 2$ in Eq. (32). Once the 3PCF multipoles have been calculated, the configuration-space shear 3PCF and aperture-mass statistics are obtained by the usual angular sum and filter integrals. This calculation has also been done with the 2D-FFTLog algorithm in Ref. [50].

For the shear example in the right panel of Fig. 1, we place a normalized thin lens at $z = 0.7$ and set its overall lensing-efficiency prefactor to unity, since it cancels from the fractional comparison. As a simplified source for H_0 , we average $\chi^{-4} B_m(\ell_1/\chi, \ell_2/\chi, \ell_3/\chi; z)$ uniformly over the opening-angle cosine with 48-point Gauss–Legendre quadrature, where χ is the comoving distance to the lens and $\ell_3 = |\ell_1 + \ell_2|$. A physical prediction instead constructs H_0 from the complete multipole coupling described above. We calculate the $M = 0$ component, for which the two Bessel orders in Eq. (44) are $(-3, -3)$, show $\theta_1 = \theta_2$, and average each angular separation over the area-weighted bin $[0.95\theta, 1.05\theta]$. The undamped input is sampled over $0.5 < \ell < 10^6$. The upper panels compare diagonal slices of the FastNBessel and tensor-Simpson results; the lower panels show ϵ_{frac} .

B. Covariances of three- and higher-point functions

A likelihood analysis requires the covariance as well as the mean of each statistic. For higher-point functions, estimating the covariance from mock catalogues is expensive, and the finite number of mocks introduces additional noise. The Gaussian contribution can instead be written as Wick contractions of power spectra and evaluated analytically. For example, the 2D-FFTLog calculation of

real-space galaxy-clustering and weak-lensing covariances in Ref. [27] was used in the covariance model for DES Year 3 [26, 51, 52]. Here we consider the Gaussian covariance of anisotropic 3PCF multipoles, which have been used in the BOSS DR12 analysis of Ref. [41] and are also relevant to future spectroscopic surveys. After the angular integrations, a typical term in the Gaussian covariance of anisotropic galaxy 3PCF multipoles contains the following radial structure. To keep the expression compact, write $(r_1, r_2) = (r_{13}, r_{23})$ and $(\lambda_1, \lambda_2) = (\ell_3, \ell_{13})$, with primed quantities referring to the second triangle. Following Eqs. (50)–(54) of Ref. [43], the radial part contains terms of the form

$$C_{\zeta\zeta} \supset \int_0^\infty \frac{dk_1}{k_1} \int_0^\infty \frac{dk_2}{k_2} \Delta_{PPP}(k_1, k_2) \times j_{\lambda_1}(k_1 r_1) j_{\lambda'_1}(k_1 r'_1) j_{\lambda_2}(k_2 r_2) j_{\lambda'_2}(k_2 r'_2) + \text{five other permutations.} \quad (45)$$

Here $C_{\zeta\zeta}$ is the covariance between two 3PCF multipoles. The r_a and r'_a are the radial separations of the two triangles, and the λ_a and λ'_a are their angular-momentum labels. The source Δ_{PPP} is shorthand for the relevant dimensionless angular multipole of $\hat{P}_g(k_1) \hat{P}_g(k_2) \hat{P}_g(k_{12})/V_s$, where $k_{12} = |\mathbf{k}_1 + \mathbf{k}_2|$, V_s is the survey volume, and $\hat{P}_g = P_g + 1/\bar{n}_g$ includes Poisson shot noise. The angular indices are suppressed in Eq. (45).

The radial integral has four Bessel functions and two integration variables, with two functions assigned to each variable. It is thus a direct $(N_B, d) = (4, 2)$ application of FastNBessel. The covariance calculation in Ref. [43] evaluates the two double-Bessel power-law integrals using Gauss hypergeometric functions. FastNBessel evaluates the same four-Bessel transform without them.

Higher N -point correlation functions lead to closely related radial kernels. In an isotropic basis, their Gaussian covariances can be assembled from a small set of such building blocks. For example, measurements of parity-odd modes of the BOSS 4PCF have been used to test parity symmetry on cosmological scales [53–55]. Primordial models can lead to the same class of radial integrals; for example, the axion-inflation trispectrum of Ref. [56] reduces to low-dimensional integrals after the angular integrations. One of the common building blocks of the Gaussian covariance is given in Eq. (38) of Ref. [57],

$$f_{\ell_1 \ell_2 \ell_3}(r_1, r_2, r_3) = \int_0^\infty \frac{k^2 dk}{2\pi^2} P(k) \times j_{\ell_1}(kr_1) j_{\ell_2}(kr_2) j_{\ell_3}(kr_3), \quad (46)$$

where k is the wavenumber magnitude and $P(k)$ is the relevant galaxy or matter power spectrum, including a shot-noise term when required. The three arguments are the two endpoint separations r_i and r'_j and the separation s between the origins of the two N -point estimators; the non-negative integers ℓ_i are their angular-momentum labels. Equation (46) is one three-Bessel building block of the complete Gaussian covariance, which combines such

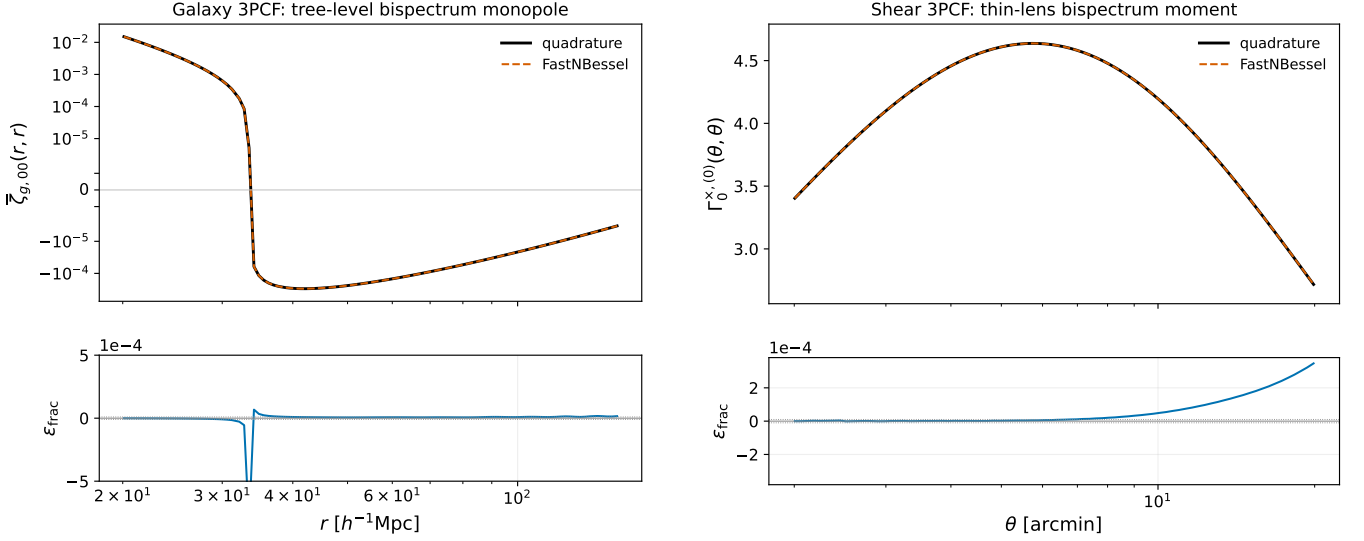


FIG. 1. Galaxy 3PCF [Eq. (42), left] and shear 3PCF [Eq. (44), right]. The upper panels compare FastNBessel with direct quadrature on diagonal configuration slices; the lower panels show the fractional difference ϵ_{frac} .

kernels through angular couplings and Wick contractions. FastNBessel evaluates Eq. (46) along a fixed-ratio ray $(r_1, r_2, r_3) = r(1, t_2, t_3)$. A full three-dimensional grid can be covered by repeating the transform for the required values of (t_2, t_3) . The volume-weighted radial bins used in the covariance calculation replace $j_\ell(kr_i)$ by the average,

$$\bar{j}_\ell(k; r_i) = \frac{\int_{r_{i,-}}^{r_{i,+}} r^2 dr j_\ell(kr)}{\int_{r_{i,-}}^{r_{i,+}} r^2 dr}, \quad (47)$$

where $r_{i,-}$ and $r_{i,+}$ are the lower and upper edges of the i th radial bin. This volume-weighted average corresponds to $D = 3$ in Sec. II C.

For the first covariance example, we isolate the displayed radial block rather than assembling the six Wick pairings and angular sums. We use an illustrative constant-density sample with $b_1 = 2$, $\bar{n}_g = 3 \times 10^{-4} h^3 \text{Mpc}^{-3}$, and $V_s = 5 (h^{-1} \text{Gpc})^3$. The isotropic galaxy spectrum entering the complete covariance is therefore $\hat{P}_g(k) = b_1^2 P_L(k, z = 0.7) + 1/\bar{n}_g$. Constant shot-noise factors generate contact terms and are normally treated separately after radial binning. The plotted ordinary integral retains the clustering part $b_1^2 P_L$ and the V_s^{-1} normalization. Its angular monopole is evaluated with 48-point Gauss–Legendre quadrature. We choose $(\lambda_1, \lambda'_1, \lambda_2, \lambda'_2) = (0, 2, 1, 1)$ and $(r_1, r'_1, r_2, r'_2) = r(1, 0.85, 1, 0.9)$, with a volume-weighted $[0.95r_i, 1.05r_i]$ bin for each separation. The input is evaluated over $10^{-5} < k < 50 h \text{Mpc}^{-1}$, with the same distant endpoint windows as the galaxy 3PCF example, and then zero-padded.

For the second covariance example, we use the matter power spectrum $P_L(k, z = 0.7)$ in Eq. (46), without a galaxy-bias or shot-noise term. We take $(\ell_1, \ell_2, \ell_3) = (0, 1, 2)$ and $(r_1, r_2, r_3) = r(1, 0.8, 1.2)$, and do not apply radial-bin averaging. The undamped source is evaluated

over $10^{-7} < k < 10^3 h \text{Mpc}^{-1}$. Raised-cosine windows are confined to $10^{-7} < k < 10^{-6}$ and $100 < k < 10^3 h \text{Mpc}^{-1}$; removing them changes the displayed integral by less than 10^{-7} . Figure 2 compares both calculations with tensor-product quadrature.

C. The spherical Fourier–Bessel bispectrum

The spherical Fourier–Bessel (SFB) basis decomposes a three-dimensional field into radial Bessel and spherical-harmonic modes. It retains radial information without dividing the survey into redshift shells, treats the curved sky, and includes the radial selection function and redshift-space distortions directly. For example, an SFB analysis was applied to the final 2dF Galaxy Redshift Survey catalogue [58]. Subsequent work developed fast treatments of the survey window [59] and forecasts for SPHEREx-like surveys [60]. The approach has also been used to study wide-angle effects in density and peculiar-velocity statistics [61–63]. We consider its application to the galaxy bispectrum. A typical radial kernel entering the SFB galaxy bispectrum is given in Eq. (70) of Ref. [64],

$$\mathcal{J}_{l_1 l_2 l_3}^{\ell L_3 L_4}(k, k_1, k_2) = \int_0^\infty r^2 dr j_\ell(kr) j_{L_3}(k_1 r) j_{L_4}(k_2 r) \times W(r) D^2(r) Z_{l_1 l_2 l_3}(k_1, k_2, r), \quad (48)$$

where $\mathcal{J}$ is a radial SFB mode-coupling kernel, r is the comoving distance from the observer, and (k, k_1, k_2) are radial wavenumbers. The indices ℓ, L_3, L_4 are the angular momenta of the three SFB modes, whereas l_1, l_2, l_3 label the Legendre templates used to decompose the second-order redshift-space kernel. The function $W(r)$ is the radial survey window, $D(r)$ is the linear growth factor, and

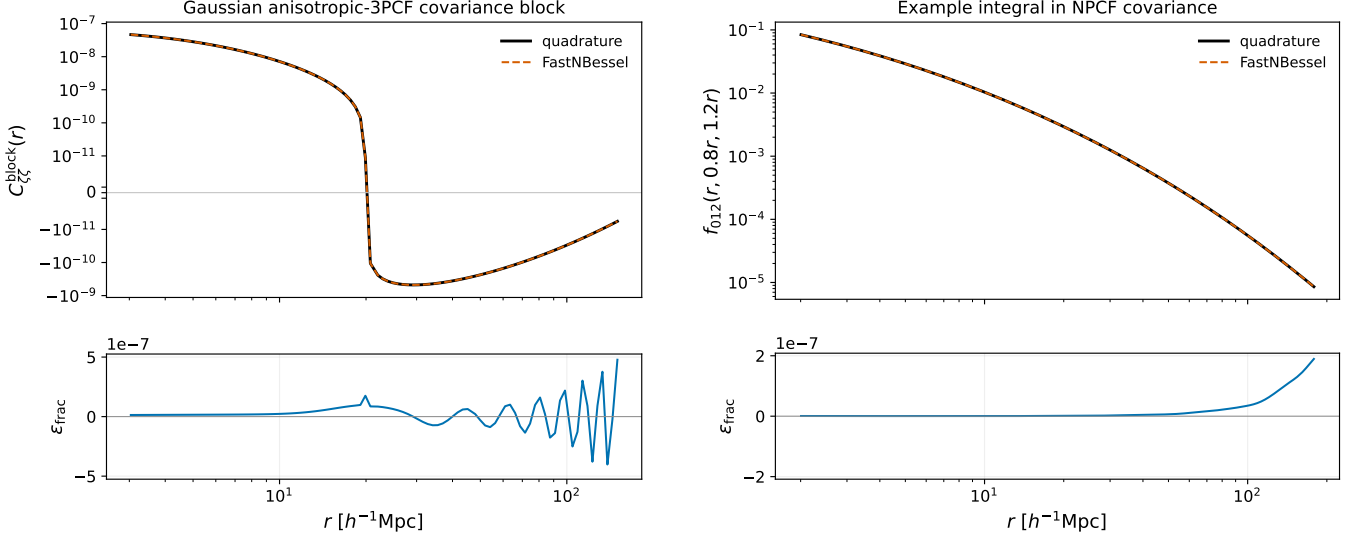


FIG. 2. Example integrals in the Gaussian covariance of an anisotropic galaxy 3PCF [Eq. (45), left] and a general galaxy N -point function [Eq. (46), right]. The lower panels show the fractional difference ϵ_{frac} .

$Z_{l_1 l_2 l_3}$ is the corresponding template coefficient containing galaxy bias, redshift-space distortions, and primordial non-Gaussianity. Equation (111) and Appendix D.1 of Ref. [64] define the complete template decomposition. At fixed $(k_1/k, k_2/k)$, Eq. (48) is a three-Bessel transform of the form in Sec. II A. As in Eq. (46), several fixed-ratio rays are required for a general (k, k_1, k_2) grid.

The SFB window provides a second useful example, given in Eq. (17) of Ref. [64],

$$\mathcal{W}_\ell(k, q) = \frac{2qk}{\pi} \int_0^\infty r^2 dr W(r) D(r) j_\ell(kr) \times [b(r, q) j_\ell(qr) - f(r) j_\ell''(qr)]. \quad (49)$$

The kernel $\mathcal{W}_\ell(k, q)$ connects a matter mode with radial wavenumber q to an observed galaxy mode with radial wavenumber k . The function $b(r, q)$ is the linear galaxy bias, while $f(r) = d \ln D(r) / d \ln a$ is the linear growth rate and a is the scale factor. The primes denote derivatives with respect to the Bessel argument qr . The derivative kernels in Sec. II C evaluate the second term without using recurrence relations or finite differencing. FastNBessel can consequently be used for both the window functions and the three-Bessel kernels that are repeatedly needed in the SFB bispectrum.

In Fig. 3, we choose a Gaussian radial selection function centered at $z = 0.7$ with width $\sigma_z = 0.18$ and normalize it to $\int r^2 dr W(r) = 1$. We use $D(0) = 1$, a bias $b(z) = 1.5 + 0.5z$, and a growth rate $f(z) = \Omega_m(z)^{0.55}$. For the bispectrum kernel we use one simple template, $Z_{l_1 l_2 l_3} = b(z)(2/7) + 0.15$, choose $(\ell, L_3, L_4) = (0, 1, 2)$, and evaluate the ray $(k, k_1, k_2) = k(1, 0.8, 1.2)$. For the window we take $\ell = 2$, set $q = 0.8k$, and include both the density and j_ℓ'' redshift-space distortion terms. For these numerical tests, we do not include shot-noise or survey-volume factors in either radial kernel.

D. Two-loop perturbation theory

Perturbative corrections extend full-shape power-spectrum predictions into the mildly nonlinear regime. One-loop models are already used in survey likelihoods, for example in the DESI DR1 analysis [65]. At two-loop order, the calculation contains several coupled momentum integrals for every external wavenumber. The two-loop matter power spectrum has been calculated, e.g., in the effective field theory of large-scale structure [66] and in standard perturbation theory with the exact Λ CDM time dependence [67]. Previous work, e.g., Refs. [17, 20], has reduced much of the two-loop calculation to low-dimensional radial transforms that can be evaluated efficiently with FFTLog, where hypergeometric functions need to be computed. Separating radial and angular dependences leads to building blocks of the form shown below, as given in Eqs. (52) and (53) of Ref. [68],

$$\mathcal{F}_{LL'L_2}(n, n'; r) = \int_0^\infty \frac{q^2 dq}{2\pi^2} F(q) j_{L_2}(qr) j_L(qn) j_{L'}(qn'), \quad (50)$$

where q is the magnitude of the inner loop momentum and r is the configuration-space separation introduced by the inverse Fourier transform. The integers L, L' , and L_2 are spherical-harmonic orders, while n and n' are radial parameters introduced by the decoupling expansion. We use $F(q)$ to denote the remaining smooth source, which contains a power of q , the linear power spectrum, and radial functions obtained at earlier stages of the calculation. Thus $\mathcal{F}_{LL'L_2}$ is one radial function entering the full two-loop result. This is another three-Bessel transform. At fixed $(n/r, n'/r)$ it can be evaluated on the entire r grid at once. The same observation applies to related loop corrections to higher-order statistics.

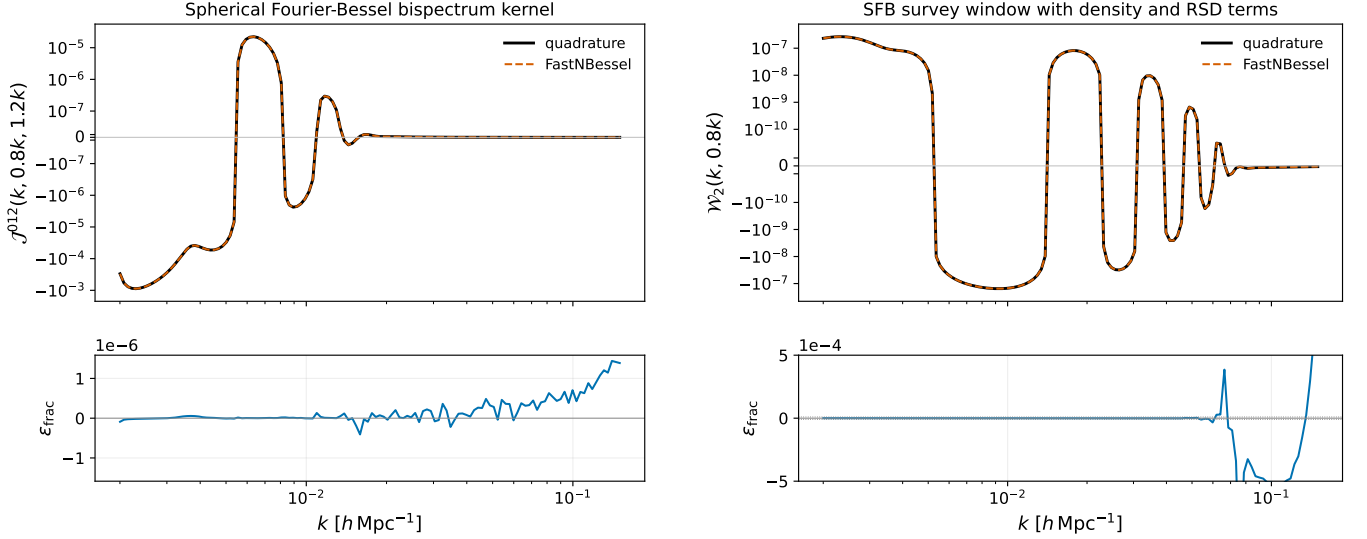


FIG. 3. SFB bispectrum radial kernel [Eq. (48), left] and density plus redshift-space distortion window kernel [Eq. (49), right], compared with direct quadrature. The lower panels show the fractional difference ϵ_{frac} .

For the example in Fig. 4, we use $F(q) = P_L(q, z = 0.7)$ over $10^{-7} < q < 10^3 h \text{ Mpc}^{-1}$. The same distant endpoint windows as in the N -point covariance example are applied before the FFT. Applying the windows changes the displayed integral by less than 10^{-7} . We evaluate the orders $(L_2, L, L') = (0, 1, 2)$ along $(r, n, n') = r(1, 0.75, 1.25)$. This is a representative radial block, not the sum of all terms in the two-loop power spectrum.

IV. DISCUSSION AND SUMMARY

A. Accuracy, speed, and parameter choices

There are two numerical grids in FastNBessel. The outer FFTLog samples $f(x)$ or $\Phi(\mathbf{x})$ in a finite logarithmic interval. Its accuracy depends on this interval, the numbers of samples N_α , the biases ν_α , the zero-padding, and the window W_s . The Mellin convolution has a separate frequency grid. Its accuracy depends on the spacing $\lambda_{\text{int},\alpha} = \lambda_\alpha / r_\eta$, where $\lambda_\alpha = 2\pi / (N_\alpha \Delta_\alpha)$, the cutoff $N_{\eta,\alpha}$, the bias split $\beta_{\alpha\alpha}$, the internal windows $W_{\eta,\alpha}$ applied to the $Q_{\alpha\alpha}$ arrays, and the zero-padding used for the linear convolution. The outer window W_s and internal windows $W_{\eta,\alpha}$ control different truncation effects and should be varied separately.

The FFTLog bias is chosen to reduce the slope of the input function. For an undifferentiated spherical Bessel function, every single-function bias must satisfy $-\ell_{\alpha\alpha} < \beta_{\alpha\alpha} < 2$ and should not be too close to a pole of the gamma functions. For a derivative of order q , the shifted argument $\beta_{\alpha\alpha} - q$ in Eq. (30) must satisfy the corresponding convergence condition. An approximately equal division of ν_α among the Bessel functions is a convenient initial choice. More Mellin modes may be needed when

two or more Bessel scale ratios $t_{\alpha\alpha}$ within a group are equal or nearly equal. At large u , the product of Bessel functions contains terms proportional to $\exp[iu \sum_a \sigma_a t_{\alpha\alpha}]$, where $\sigma_a = \pm 1$. If one of the signed sums vanishes, the corresponding term does not oscillate and its Mellin transform decreases more slowly. In this case both $N_{\eta,\alpha}$ and the convolution padding should be increased. We evaluate the complex gamma-function ratios using log-gamma functions to avoid overflow (see Appendix C).

To reduce the ringing effect in the FFT, we evaluate or extrapolate the physical input to a broad interval, taper only its distant endpoints, and add zeros while preserving the logarithmic spacing. A finer quadrature grid is needed to resolve the same rapidly oscillating tails directly. Increasing the internal Mellin oversampling from two to four was the leading improvement for the triple-Bessel examples. The internal sampling affects the cost of constructing the cosmology-independent Mellin kernels, but not the outer FFTLog once those kernels have been computed. For axis α , the convolution length is $L_\alpha = p_\alpha(N_{\eta,\alpha} - 1) + 1$, so kernel construction requires $\mathcal{O}[(p_\alpha + 1)L_\alpha \log L_\alpha]$ operations and $\mathcal{O}(L_\alpha)$ memory. At fixed $N_{\eta,\alpha}$, changing r_η changes the frequency spacing but not this asymptotic cost. In practice, covering all outer modes requires approximately $N_{\eta,\alpha} \gtrsim 1 + r_\eta N_\alpha / p_\alpha$, so increasing r_η generally also increases the kernel time and memory. Once the Bessel orders, ratios, grids, and biases are fixed, the kernels can be stored and reused, leaving the cost of each additional transform unchanged. The convergence and timing tests in Appendix E show how these choices affect the numerical results.

For a tensor-product quadrature rule, define the one-

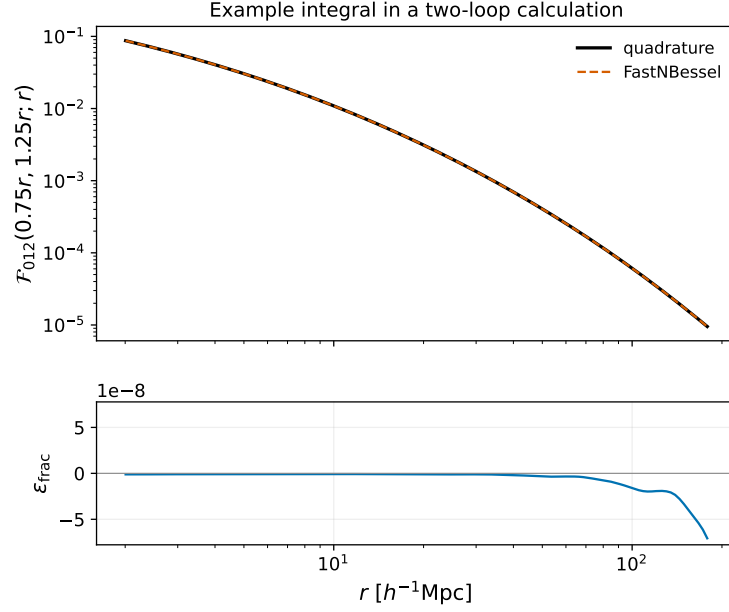


FIG. 4. Example integral in a two-loop power spectrum calculation [Eq. (50)], compared with direct quadrature. The lower panel shows the fractional difference ϵ_{frac} .

dimensional Bessel products

$$J_\alpha(n, q) = \prod_{a=1}^{p_\alpha} j_{\ell_{\alpha a}}(t_{\alpha a} x_{\alpha q} y_{\alpha n}). \quad (51)$$

With N_x integration samples and N_y output samples in each dimension, the discrete transform has the form

$$I_n \simeq \sum_{q_1 \dots q_d} \Phi_q \prod_{\alpha=1}^d w_{\alpha q_\alpha} J_\alpha(n_\alpha, q_\alpha), \quad (52)$$

where $w_{\alpha q}$ are the quadrature weights. A brute-force implementation loops over N_y^d output points and N_x^d integration nodes and evaluates all $N_B = \sum_\alpha p_\alpha$ Bessel functions inside the innermost loop, giving $\mathcal{O}(N_B N_x^d N_y^d)$ operations.

The tensor-Simpson references in this paper are evaluated only at the plotted output points. For one output point $\mathbf{n}$, forming all of the one-dimensional Bessel products in Eq. (51) requires $N_B N_x$ operations. Multiplying these products by Φ_q on the N_x^d integration grid and summing over each axis requires another $d N_x^d$ operations. Repeating this calculation independently at all N_y^d output points would therefore cost

$$\mathcal{O}\{N_y^d [N_B N_x + d N_x^d]\}. \quad (53)$$

On a full Cartesian output grid, much of this work can be reused. The matrices $J_\alpha(n, q)$ are first calculated and stored at a cost $\mathcal{O}(N_B N_x N_y)$. Equation (52) is then contracted one integration axis at a time. After $\alpha - 1$ contractions, the intermediate array has $N_y^{\alpha-1} N_x^{d-\alpha+1}$

elements, and the next contraction costs $N_y^\alpha N_x^{d-\alpha+1}$ operations. The total cost is therefore

$$\mathcal{O}\left[N_B N_x N_y + \sum_{\alpha=1}^d N_y^\alpha N_x^{d-\alpha+1}\right]. \quad (54)$$

This contraction is unavailable without modification for scattered output points or independently adapted integration grids, but it is the appropriate quadrature comparison on a complete Cartesian grid.

FastNBessel instead requires

$$\mathcal{O}\left[N_{\text{grid}} \sum_{\alpha=1}^d \log N_\alpha + \sum_{\alpha=1}^d (p_\alpha + 1) L_\alpha \log L_\alpha\right] \quad (55)$$

operations and $\mathcal{O}(N_{\text{grid}} + \sum_\alpha L_\alpha)$ memory, where L_α is the padded length of a one-dimensional Mellin convolution. For equal grids and fixed N_B and d , the leading time scaling is $\mathcal{O}(N^{2d})$ for independent point-by-point quadrature, $\mathcal{O}(d N^{d+1})$ for the optimized contraction in Eq. (54), and $\mathcal{O}(d N^d \log N)$ for FastNBessel. The optimized quadrature and FastNBessel calculations both require $\mathcal{O}(N^d)$ leading memory space; forming all integration-node-output-point pairs simultaneously would instead require $\mathcal{O}(N^{2d})$ memory space.

B. Summary

In this paper we have introduced FastNBessel, a novel FFTLog algorithm for transforms containing an arbitrary number of Bessel functions (Sec. II). We have applied the method to galaxy and shear three-point functions,

covariance kernels, SFB kernels, and an integral entering a two-loop power spectrum calculation (Sec. III). The analytic tests in Appendix E give median absolute fractional differences between 10^{-8} and 3×10^{-7} . For equal grids and fixed N_B and d , the leading cost is $\mathcal{O}(dN^d \log N)$, compared with $\mathcal{O}(N^{2d})$ for independent point-by-point quadrature and $\mathcal{O}(dN^{d+1})$ for optimized tensor contraction; the leading memory space is $\mathcal{O}(N^d)$ (Sec. IV A).

The efficiency of FFTLog follows from the dilation symmetry of the Bessel kernel, which becomes translational symmetry in logarithmic coordinates. When several Bessel functions share one integration variable, the Mellin product theorem reduces their transform to a convolution of single-Bessel gamma-function kernels, which can be evaluated with FFTs (Sec. II A). For several integration variables, a multidimensional FFTLog separates the power-law integrals and the same one-dimensional Mellin construction is applied to each variable (Sec. II B). The resulting kernels are independent of the input function and can be reused. No hypergeometric functions are required, and adding Bessel functions does not change the form of the algorithm. Derivatives, bin averages, and cylindrical Bessel functions are included in Secs. II C and II D. Analytic kernels for other Bessel families are given in Sec. II E; their implementation and numerical tests are left to future work.

The algorithm and the examples presented here are publicly available as the FastNBessel Python package at <https://github.com/xfangcosmo/FastNBessel>.

ACKNOWLEDGMENTS

The author used GPT-5.6 to significantly accelerate code development and numerical testing, which have traditionally been the most time-consuming parts of the research process. GPT-5.6 was also used to improve the writing of this manuscript.

DATA AVAILABILITY

The source code, benchmark and plotting scripts, and data used to make the figures are available in the public FastNBessel repository. The fiducial input spectra and kernels used in Sec. III are distributed as text tables together with the scripts that generate them. This work does not use observational data.

Appendix A: Useful Mellin-transform identities

We use the following definitions for the Mellin transform and its inverse [Sec. 2.5(i) of Ref. [31]; see also Ref. [32]]:

$$F(z) = \mathfrak{M}[f](z) = \int_0^\infty \frac{du}{u} u^z f(u), \quad (\text{A1})$$

$$f(u) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} dz u^{-z} F(z), \quad (\text{A2})$$

where the vertical contour $\text{Re } z = c$ is inside the fundamental strip of F . Several identities used in the calculation are

$$\mathfrak{M}[f(au)](z) = a^{-z} F(z), \quad a > 0, \quad (\text{A3})$$

$$\mathfrak{M}[u^\rho f(u)](z) = F(z + \rho), \quad (\text{A4})$$

$$\mathfrak{M}[(\ln u)^q f(u)](z) = \frac{d^q}{dz^q} F(z), \quad (\text{A5})$$

$$\mathfrak{M}[f(u^{-1})](z) = F(-z), \quad (\text{A6})$$

$$\mathfrak{M}[f^{(q)}](z) = (-1)^q \frac{\Gamma(z)}{\Gamma(z-q)} F(z-q). \quad (\text{A7})$$

Equation (A7) assumes that the boundary terms vanish in each integration by parts. The gamma-function ratio is the finite product $(z-1) \cdots (z-q)$. We evaluate this finite product directly when the individual gamma functions are singular.

The product, multiplicative-convolution, and Parseval identities below are standard Mellin-transform results [31, 32]. The Mellin transform of a product is

$$\mathfrak{M}[fg](z) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} dw F(w) G(z-w), \quad (\text{A8})$$

where the contour lies in the common fundamental strip. Applying this identity repeatedly gives Eq. (11). The corresponding multiplicative convolution is defined by

$$(f \star_M g)(u) = \int_0^\infty \frac{dv}{v} f(v) g(u/v), \quad (\text{A9})$$

and its Mellin transform is

$$\mathfrak{M}[f \star_M g](z) = F(z) G(z). \quad (\text{A10})$$

The Parseval identity is

$$\int_0^\infty \frac{du}{u} f(u) g(u) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} dz F(z) G(-z), \quad (\text{A11})$$

where the contour is chosen so that both sides converge. For a real function f ,

$$F(z^*) = F(z)^*, \quad F(\beta - i\eta) = F(\beta + i\eta)^*, \quad (\text{A12})$$

which is analogous to the Hermitian symmetry of a Fourier transform.

The Mellin transforms of the cylindrical and spherical Bessel functions used in this paper follow from the standard power-Bessel integral [Eq. (10.22.43) of Ref. [31]; see also Ref. [32]] and are

$$\int_0^\infty du u^{z-1} J_n(u) = 2^{z-1} \frac{\Gamma[(n+z)/2]}{\Gamma[(2+n-z)/2]}, \quad (\text{A13})$$

valid for $-n < \text{Re } z < 3/2$, and

$$\int_0^\infty du u^{z-1} j_\ell(u) = \frac{\sqrt{\pi}}{4} 2^z \frac{\Gamma[(\ell+z)/2]}{\Gamma[(3+\ell-z)/2]}. \quad (\text{A14})$$

The second expression is valid for $-\ell < \text{Re } z < 2$. Equation (A14) is the same as Eq. (6). Its scaled and differentiated forms follow from Eqs. (A3) and (A7). These upper limits apply to the conditionally convergent integrals; absolute convergence requires a smaller fundamental strip.

Appendix B: Discrete Fourier conventions and normalization

For the forward transform

$$\tilde{c}_s = \sum_{q=0}^{N-1} a_q e^{-2\pi i s q / N} \quad (\text{B1})$$

we define

$$\text{IFFT}[A]_n = \frac{1}{N} \sum_{s=-N/2}^{N/2-1} A_s e^{+2\pi i s n / N}. \quad (\text{B2})$$

The transform in Eq. (21) has a negative Fourier phase in the output index. Reversing the spectral indices, or using Hermitian symmetry for real data, gives Eq. (22). The IFFT supplies the factor $1/N$, while $(\lambda/2\pi)^{p-1}$ comes from the Mellin quadrature. There are no additional factors of N .

The same convention is used for each of the d dimensions. If the arrays before the transforms are real, $c_{-s} = c_s^*$ and $K_{\alpha,-s} = K_{\alpha s}^*$, which produces Eq. (29). We do not need to store the full multidimensional kernel, since the one-dimensional arrays $K_{\alpha s}$ can be broadcast over the coefficient array.

As a separate check of the normalization, the common- x integral can be written as a diagonal p -dimensional transform. With the measure $d \ln x$, the discrete form of $f(x_1) \prod_{a=2}^p \delta_D(\ln x_a - \ln x_1)$ is

$$f_{q_1 \dots q_p}^{\text{diag}} = \frac{f(x_{q_1})}{\Delta^{p-1}} \prod_{a=2}^p \delta_{q_a q_1}. \quad (\text{B3})$$

The identity

$$\frac{1}{N^p \Delta^{p-1}} = \frac{1}{N} \left(\frac{\lambda}{2\pi} \right)^{p-1} \quad (\text{B4})$$

reproduces the normalization in Eq. (17).

Appendix C: Convergence conditions

Near the origin, the Bessel product scales as

$$\prod_a j_{\ell_a}(t_a u) \sim u^{\sum_a \ell_a}.$$

Therefore, convergence at the lower limit requires

$$\nu_\alpha > -\sum_{a=1}^{p_\alpha} \ell_{\alpha a}. \quad (\text{C1})$$

At large u , the product is a sum of phase channels of the form

$$u^{-p_\alpha} \exp\left(iu \sum_a \sigma_a t_{\alpha a}\right), \quad \sigma_a = \pm 1. \quad (\text{C2})$$

The large- u integral is absolutely convergent if $\nu_\alpha < p_\alpha$. If all the signed frequencies are nonzero, the oscillations extend the condition to $\nu_\alpha < p_\alpha + 1$. If one signed sum vanishes, the stricter condition is required.

Each Mellin contour associated with an individual Bessel factor must lie in its fundamental strip, $-\ell_{\alpha a} < \beta_{\alpha a} < 2$. The large-frequency behavior

$$|g_\ell(\beta + i\eta)| \sim \text{const. } |\eta|^{\beta-3/2} \quad (\text{C3})$$

shows that degenerate ratios or a poor choice of the bias split can give a slowly decreasing Mellin kernel. The individual fundamental-strip conditions do not by themselves guarantee absolute convergence of the convolution: several Mellin frequencies can become large while canceling in the sum that fixes the outer frequency. Let A contain $m = |A|$ such frequencies. The cancellation constraint leaves $m-1$ independent large-frequency directions. Writing their common scale as ρ , a radial shell in this space has measure $\rho^{m-2} d\rho$. The product of the m asymptotic kernels contributes $\rho^{\sum_{a \in A} \beta_{\alpha a} - 3m/2}$, so this part of the convolution behaves as

$$\int_0^\infty d\rho \rho^{\sum_{a \in A} \beta_{\alpha a} - m/2 - 2}. \quad (\text{C4})$$

Absolute convergence requires the exponent to be smaller than -1 and hence

$$\sum_{a \in A} \beta_{\alpha a} < 1 + \frac{|A|}{2} \quad \text{for every } A \subseteq \{1, \dots, p_\alpha\}, \quad |A| \geq 2. \quad (\text{C5})$$

This condition is not necessary. Outside this region, the Mellin integral may still converge conditionally or be defined by analytic continuation, but a truncated convolution can converge slowly and contain large cancellations.

Appendix D: Derivatives, bin averages, and J_n kernels

Integrating by parts once gives

$$\int_0^\infty du u^{z-1} j'_\ell(u) = -(z-1) \int_0^\infty du u^{z-2} j_\ell(u), \quad (\text{D1})$$

when the boundary term vanishes. Repeating this step gives $(-1)^q \Gamma(z) U_\ell(z - q) / \Gamma(z - q)$, as used in Eq. (30). If the derivative is with respect to x or y rather than the Bessel argument, it also gives powers of the other variable and t_a . These factors can be written explicitly or included in the input function.

For the bin average, we change variables to $w = vu$ and exchange the order of integration:

$$\int_0^\infty \frac{du}{u} u^z \bar{j}_\ell(u; a, b) = \frac{D}{b^D - a^D} \int_a^b dv v^{D-1-z} U_\ell(z) \quad (\text{D2})$$

$$= B_D(z; a, b) U_\ell(z). \quad (\text{D3})$$

At $z = D$, we use the limit

$$B_D(D; a, b) = \frac{D \ln(b/a)}{b^D - a^D}. \quad (\text{D4})$$

Inserting $J_n(u) = \sqrt{2u/\pi} j_{n-1/2}(u)$ into Eq. (6) gives Eq. (33), with the conditional fundamental strip $-n < \text{Re } z < 3/2$.

Appendix E: Numerical validation

We compare FastNbessel and direct-quadrature calculations using smoothly damped inputs whose contributions beyond the integration ranges are negligible.

1. Reference calculations

For the one-dimensional tests, we use composite Simpson integration in $\ln k$ with 32769 points over the same $10^{-7} \leq k \leq 10^3$ interval as the FastNbessel calculation. For the two-dimensional tests, we use 501 points per axis over $10^{-6} \leq k_\alpha \leq 10^3$; for the three-dimensional test, we use 193 points per axis over $10^{-7} \leq k_\alpha \leq 30$. We precompute the Bessel functions and contract these arrays with tensor-product Simpson weights. The sampling is increased until the reference values no longer change at the accuracy quoted below.

For each test, we record the physical integration range, the outer grid and biases, the Mellin spacing and cutoff, the padding and windows, the quadrature sampling or tolerance, the hardware and software versions, and the timing procedure. We compare the two methods only after each has converged independently. For a separable input, Eq. (24) must reduce to a product of one-dimensional transforms, which provides a useful check of the multidimensional normalization. The reference calculation uses direct Bessel evaluations and Simpson weights and does not call the transform code.

2. One-dimensional multiple-Bessel integrals

The one-dimensional tests use

$$I_p(r) = \int_0^\infty \frac{dk}{k} k^{m_p} e^{-k^2/2} \prod_{a=1}^p j_{\ell_a}(t_a k r), \quad (\text{E1})$$

where the orders, ratios, and powers are listed in Table III. The $p = 2$ and $p = 3$ examples are also used in the package tests. The $p = 4$ and $p = 5$ examples require longer Mellin convolutions, but the outer FFTLog calculation is unchanged.

The four integrals are

$$I_2(r) = \int_0^\infty \frac{dk}{k} k^2 e^{-k^2/2} j_0(kr) j_2(0.83kr), \quad (\text{E2})$$

$$I_3(r) = \int_0^\infty \frac{dk}{k} k^3 e^{-k^2/2} j_0(kr) j_1(0.8kr) j_2(1.3kr), \quad (\text{E3})$$

$$I_4(r) = \int_0^\infty \frac{dk}{k} k^4 e^{-k^2/2} j_0(kr) j_1(0.73kr) \times j_2(1.11kr) j_3(1.41kr), \quad (\text{E4})$$

$$I_5(r) = \int_0^\infty \frac{dk}{k} k^5 e^{-k^2/2} j_0(kr) j_1(0.69kr) j_2(0.91kr) \times j_3(1.17kr) j_4(1.43kr). \quad (\text{E5})$$

The first example is the simplest case that requires a Mellin convolution. The last two test the same algorithm beyond the three Bessel functions supported by the specialized Levin method of Ref. [12].

The FastNbessel settings for all four cases are listed in Table IV. They use the same 512-point input grid $10^{-7} \leq k \leq 10^3$ and the corresponding 512-point output grid $10^{-2} \leq r \leq 10^8$. Figure 5 shows the range $0.03 \leq r \leq 30$.

3. Multidimensional integrals

For the multidimensional examples we use the input function

$$\Phi_d(\mathbf{k}) = \left(\prod_{\alpha=1}^d k_\alpha^2 \right) \exp \left(-\frac{1}{2} \sum_{\alpha} k_\alpha^2 \right) \times \left[1 + 0.2 \exp \left(-\frac{1}{2\sigma^2} \sum_{\alpha < \beta} \ln^2 \frac{k_\alpha}{k_\beta} \right) \right], \quad (\text{E6})$$

with $\sigma = 0.8$. The logarithmic-ratio term makes the input nonseparable, so the examples test both the multidimensional FFTLog and the one-dimensional Bessel kernels.

We first consider the 2D-FFTLog limit, with one Bessel function for each integration variable,

$$T_{2,2}(r_1, r_2) = \int \frac{dk_1}{k_1} \frac{dk_2}{k_2} \Phi_2(k_1, k_2) j_0(k_1 r_1) j_2(k_2 r_2). \quad (\text{E7})$$

TABLE III. One-dimensional accuracy tests in Eq. (E1).

p	$(\ell_1, \dots, \ell_p)$	$(t_1, \dots, t_p)$	m_p
2	(0, 2)	(1, 0.83)	2
3	(0, 1, 2)	(1, 0.8, 1.3)	3
4	(0, 1, 2, 3)	(1, 0.73, 1.11, 1.41)	4
5	(0, 1, 2, 3, 4)	(1, 0.69, 0.91, 1.17, 1.43)	5

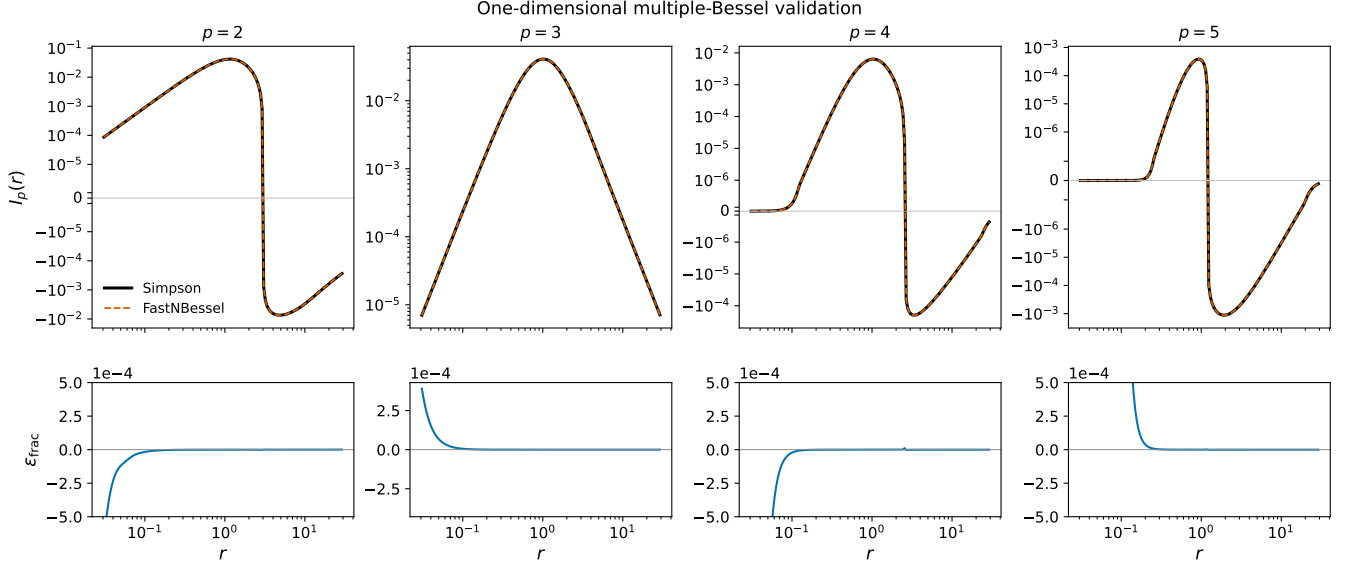


FIG. 5. Accuracy of the one-dimensional tests in Eqs. (E2)–(E5). The lower panels show the fractional difference ϵ_{frac} . The median values of $|\epsilon_{\text{frac}}|$ for $p = 2, 3, 4, 5$ are 2.5×10^{-7} , 7.0×10^{-8} , 1.5×10^{-7} , and 1.2×10^{-7} , respectively.

The four-Bessel, two-dimensional test is

$$T_{4,2}(r_1, r_2) = \int \frac{dk_1}{k_1} \frac{dk_2}{k_2} \Phi_2(k_1, k_2) j_0(k_1 r_1) j_1(0.8 k_1 r_1) \times j_0(k_2 r_2) j_2(1.3 k_2 r_2). \quad (\text{E8})$$

It uses a two-dimensional FFTLog and two double-Bessel Mellin convolutions. On the radial range used for the test, comparison with a 501×501 tensor-Simpson calculation gives a median absolute fractional difference of 1.1×10^{-7} and a maximum of 4.5×10^{-5} . For a separable input, the multidimensional result agrees with the product of the one-dimensional transforms to floating-point precision.

Finally, the six-Bessel, three-dimensional test is

$$T_{6,3}(\mathbf{r}) = \int \prod_{\alpha=1}^3 \frac{dk_{\alpha}}{k_{\alpha}} \Phi_3(k_1, k_2, k_3) j_0(k_1 r_1) j_1(0.8 k_1 r_1) \times j_1(k_2 r_2) j_2(1.2 k_2 r_2) j_0(k_3 r_3) j_2(0.9 k_3 r_3). \quad (\text{E9})$$

Direct quadrature is evaluated at a selected set of points because of its computational cost. For this most demanding example, we use 160 samples per FFTLog axis, an internal Mellin oversampling factor of four, and 193 samples per axis for the tensor-Simpson reference. The

shallower total bias $\nu_{\alpha} = 0.5$ suppresses the low-frequency periodic image that otherwise contaminates the small- r tail.

For completeness, Table IV collects the full outer-grid and Mellin-grid settings of all accuracy tests. The intervals and sample counts are per dimension.

4. Performance and convergence

We compare the wall-clock time of FastNBessel and tensor-Simpson quadrature with both methods restricted to one thread and the same fixed physical integration range. Starting from a coarse grid, we refine the Simpson grid until

$$\epsilon_Q = \max_i \frac{|Q_i^{(M)} - Q_i^{(M')}|}{\max(|Q_i^{(M)}|, 10^{-5} Q_{\max}^{(M)})} < 10^{-5}, \quad (\text{E10})$$

$$Q_{\max}^{(M)} = \max_i |Q_i^{(M)}|,$$

where $M' < M$ denotes the preceding, coarser grid. The floor in the denominator replaces a pointwise relative test by an absolute test within 10^{-5} of the peak, so points

TABLE IV. FastNBessel numerical parameters for the analytic validation examples. A semicolon in the β column separates integration variables.

Case	input interval	output interval	$N_x = N_y$	$\{\nu_\alpha\}$	$\{\beta_{\alpha a}\}$	(r_η, N_η)
I_2	$k : 10^{-7}-10^3$	$r : 10^{-2}-10^8$	512	1	(0.5, 0.5)	(2, 16385)
I_3	$k : 10^{-7}-10^3$	$r : 10^{-2}-10^8$	512	1.2	(0.4, 0.4, 0.4)	(2, 16385)
I_4	$k : 10^{-7}-10^3$	$r : 10^{-2}-10^8$	512	1.6	(0.4, 0.4, 0.4, 0.4)	(2, 32769)
I_5	$k : 10^{-7}-10^3$	$r : 10^{-2}-10^8$	512	2	(0.4, 0.4, 0.4, 0.4, 0.4)	(2, 32769)
$T_{2,2}$	$k_{1,2} : 10^{-6}-10^3$	$r_{1,2} : 10^{-2}-10^7$	128	(1, 1)	(1); (1)	—
$T_{4,2}$	$k_{1,2} : 10^{-6}-10^3$	$r_{1,2} : 10^{-2}-10^7$	128	(1, 1)	(0.5, 0.5); (0.5, 0.5)	(2, 8193)
$T_{6,3}$	$k_{1,2,3} : 10^{-7}-30$	$r_{1,2,3} : 0.02-6 \times 10^6$	160	(0.5, 0.5, 0.5)	(0.25, 0.25); (0.25, 0.25); (0.25, 0.25)	(4, 32769)

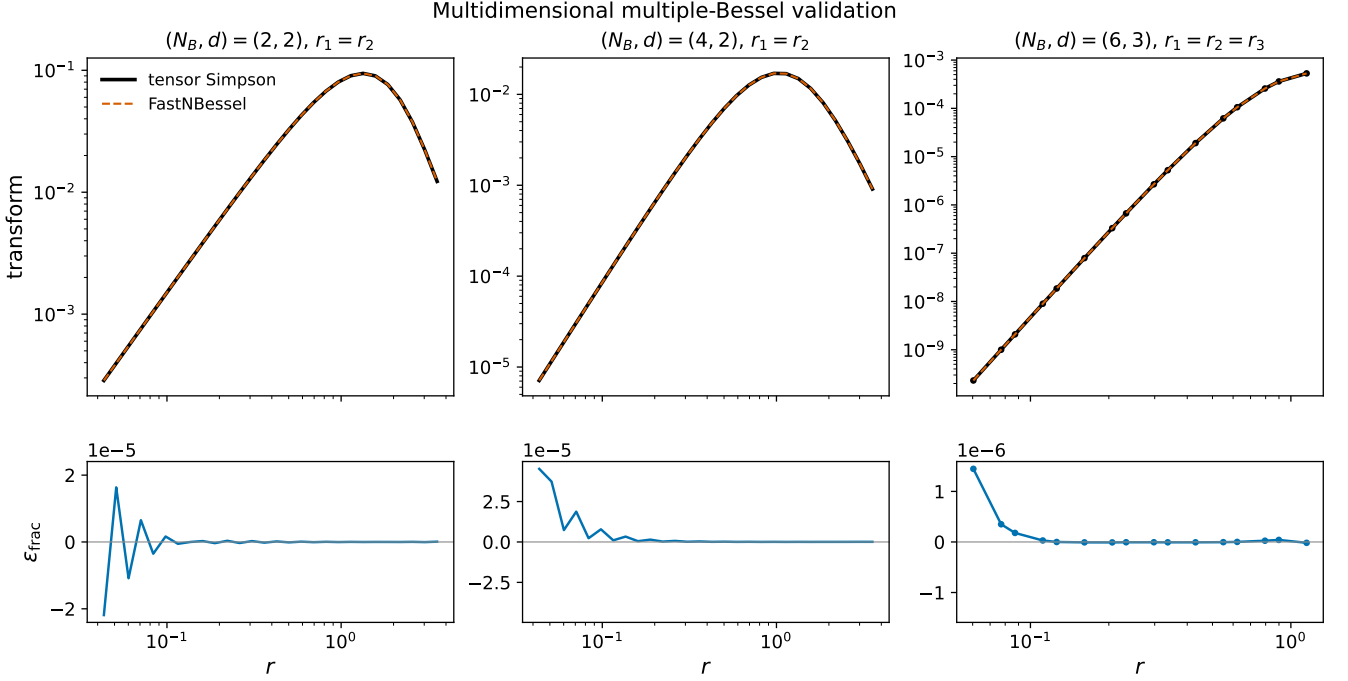


FIG. 6. Accuracy of the multidimensional tests in Eqs. (E7)–(E9). FastNBessel is compared with independent tensor-product quadrature along $r_1 = r_2$ or $r_1 = r_2 = r_3$. The median absolute fractional differences $|\epsilon_{\text{frac}}|$ are 2.1×10^{-7} , 1.1×10^{-7} , and 1.2×10^{-8} for $(N_B, d) = (2, 2)$, $(4, 2)$, and $(6, 3)$, respectively; the corresponding maxima are 2.2×10^{-5} , 4.5×10^{-5} , and 1.5×10^{-6} .

where the reference value is close to zero do not control the stopping condition. The reported quadrature time is cumulative over all successively refined grids evaluated before the criterion is satisfied. Every reported FastNBessel timing includes construction of the Mellin kernels. The benchmark was run on a MacBook Pro (Mac15,6) with an Apple M3 Pro chip, 18 GB of memory, and macOS 26.2; the numerical-library thread limits and the FFT worker count were set to one. For the multidimensional quadrature curves, the code first forms the one-dimensional Bessel products and then applies tensor-Simpson quadrature at one output point. We measure this converged one-point cost and multiply it by the number of points in the full output grid. The plotted curves are therefore estimates for independent point-by-point evaluation and do not use the full-grid tensor contraction in Eq. (54).

The left panel of Fig. 7 uses the one-dimensional

intervals and biases in Table IV, with $N_x = N_y = 64, 128, 256, 512$ and the tabulated r_η and N_η held fixed. The tensor-Simpson timing uses the same $10^{-7} \leq k \leq 10^3$ interval and $N_y = N_x$ output points from $r = 0.03$ to 30. The convergence test selects 4097 integration samples for $p = 2$ and 8193 for $p = 3, 4, 5$ at $N_y = 512$. The middle panel uses $10^{-6} \leq k_\alpha \leq 10^2$, $y_{\alpha 0} = 0.02$, $\nu_\alpha = 1$, $\beta_{\alpha a} = 0.5$, $r_\eta = 2$, and $N_\eta = 4097$. For $(N_B, d) = (4, 2)$ we take $N_\alpha = 24, 32, 48, 64$, while for $(N_B, d) = (6, 3)$ we take $N_\alpha = 12, 16, 20, 24$. The converged Simpson grids have 257 and 193 samples per integration axis, respectively. The right panel uses the I_2 outer grid with $N_x = N_y = 512$, $\nu = 1$, $(\beta_1, \beta_2) = (0.5, 0.5)$, and $r_\eta = 2$, while varying $N_\eta = 2049, 4097, 8193, 16385, 32769$.

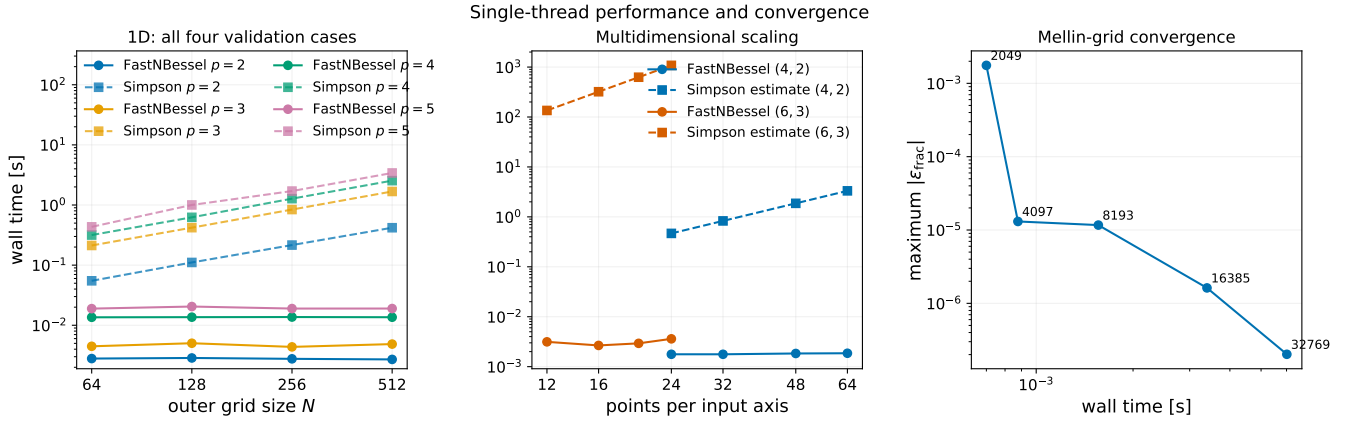


FIG. 7. Performance and convergence of FastNBessel compared with direct quadrature. All timings are restricted to one thread and include kernel construction or Simpson refinement through the criterion in Eq. (E10). Lines in the center panel labeled “Simpson estimate” extrapolate a measured one-point quadrature time to independent evaluation of the full output grid; they do not use full-grid tensor contractions. The right panel varies the Mellin-grid length for the $p = 2$ test.

-
- [1] DESI Collaboration, A. Aghamousa, *et al.*, The desi experiment part i: Science, targeting, and survey design, arXiv e-prints (2016), [arXiv:1611.00036 \[astro-ph.IM\]](#).
- [2] R. Laureijs *et al.*, Euclid definition study report, arXiv e-prints (2011), [arXiv:1110.3193 \[astro-ph.CO\]](#).
- [3] LSST Dark Energy Science Collaboration, R. Mandelbaum, *et al.*, The lsst dark energy science collaboration science requirements document, arXiv e-prints (2018), [arXiv:1809.01669 \[astro-ph.CO\]](#).
- [4] O. Doré, J. Bock, M. Ashby, P. Capak, A. Cooray, R. de Putter, T. Eifler, N. Flagey, Y. Gong, S. Habib, K. Heitmann, C. Hirata, W.-S. Jeong, R. Katti, P. Korngut, E. Krause, D.-H. Lee, D. Masters, P. Mauskopf, G. Melnick, B. Mennesson, H. Nguyen, K. Öberg, A. Pullen, A. Raccanelli, R. Smith, Y.-S. Song, V. Toll, S. Unwin, T. Venumadhav, M. Viero, M. Werner, and M. Zemcov, Cosmology with the SPHEREx All-Sky Spectral Survey, arXiv e-prints, arXiv:1412.4872 (2014), [arXiv:1412.4872 \[astro-ph.CO\]](#).
- [5] D. Spergel *et al.*, Wide-field infrared survey telescope—astrophysics focused telescope assets wfirst-afta 2015 report, arXiv e-prints (2015), [arXiv:1503.03757 \[astro-ph.IM\]](#).
- [6] K. N. Abazajian *et al.*, Cmb-s4 science book, first edition, arXiv e-prints (2016), [arXiv:1610.02743 \[astro-ph.CO\]](#).
- [7] F. Bernardeau, S. Colombi, E. Gaztanaga, and R. Scoccimarro, Large-scale structure of the universe and cosmological perturbation theory, *Physics Reports* **367**, 1 (2002), [arXiv:astro-ph/0112551](#).
- [8] O. H. E. Philcox and M. M. Ivanov, The BOSS DR12 Full-Shape Cosmology: Λ CDM Constraints from the Large-Scale Galaxy Power Spectrum and Bispectrum Monopole, *Physical Review D* **105**, 043517 (2022), [arXiv:2112.04515 \[astro-ph.CO\]](#).
- [9] A. Chudaykin, M. M. Ivanov, and O. H. E. Philcox, Reanalyzing desi dr1: 1. Λ CDM constraints from the power spectrum and bispectrum, arXiv e-prints (2025), [arXiv:2507.13433 \[astro-ph.CO\]](#).
- [10] C. Guandalin, P. Bansal, P. Carrilho, A. Aviles, M. S. Wang, M. Pellejero-Ibañez, A. Sarma, J. Swanson, M. Bonici, F. Beutler, A. de Mattia, and H.-J. Seo, Cosmological inference from a joint DESI DR1 full-shape power spectrum and bispectrum analysis, arXiv e-prints, arXiv:2610.01836 (2026), [arXiv:2610.01836 \[astro-ph.CO\]](#).
- [11] D. Levin, Fast integration of rapidly oscillatory functions, *Journal of Computational and Applied Mathematics* **67**, 95 (1996).
- [12] R. Reischke, pylevin: efficient numerical integration of integrals containing up to three bessel functions, *Journal of Open Source Software* **10**, 8618 (2025), [arXiv:2502.12142 \[math.NA\]](#).
- [13] J. D. Talman, Numerical fourier and bessel transforms in logarithmic variables, *Journal of Computational Physics* **29**, 35 (1978).
- [14] A. J. S. Hamilton, Uncorrelated modes of the non-linear power spectrum, *Monthly Notices of the Royal Astronomical Society* **312**, 257 (2000), [arXiv:astro-ph/9905191](#).
- [15] M. Schmittfull, Z. Vlah, and P. McDonald, Fast large scale structure perturbation theory using one-dimensional fast Fourier transforms, *Physical Review D* **93**, 103528 (2016), [arXiv:1603.04405 \[astro-ph.CO\]](#).
- [16] J. E. McEwen, X. Fang, C. M. Hirata, and J. A. Blazek, Fast-pt: a novel algorithm to calculate convolution integrals in cosmological perturbation theory, *Journal of Cosmology and Astroparticle Physics* **2016** (09), 015, [arXiv:1603.04826](#).
- [17] M. Schmittfull and Z. Vlah, Reducing the two-loop large-scale structure power spectrum to low-dimensional, radial integrals, *Physical Review D* **94**, 103530 (2016), [arXiv:1609.00349 \[astro-ph.CO\]](#).
- [18] X. Fang, J. A. Blazek, J. E. McEwen, and C. M. Hirata, Fast-pt ii: an algorithm to calculate convolution integrals of general tensor quantities in cosmological perturbation theory, *Journal of Cosmology and Astroparticle Physics* **2017** (02), 030, [arXiv:1609.05978](#).
- [19] V. Assassi, M. Simonović, and M. Zaldarriaga, Efficient evaluation of angular power spectra and bispectra, *Journal of Cosmology and Astroparticle Physics* **2017** (11), 054, [arXiv:1705.05022](#).
- [20] M. Simonović, T. Baldauf, M. Zaldarriaga, J. J. M. Carrasco, and J. A. Kollmeier, Cosmological Perturbation Theory Using the FFTLog: Formalism and Connection to QFT Loop Integrals, *Journal of Cosmology and Astroparticle Physics* **2018** (04), 030, [arXiv:1708.08130 \[astro-ph.CO\]](#).
- [21] J. A. Blazek, N. MacCrann, M. A. Troxel, and X. Fang, Beyond linear galaxy alignments, *Physical Review D* **100**, 103506 (2019), [arXiv:1708.09247 \[astro-ph.CO\]](#).
- [22] H. S. Grasshorn Gebhardt and D. Jeong, Fast and accurate computation of projected two-point functions, *Physical Review D* **97**, 023504 (2018), [arXiv:1709.02401](#).
- [23] N. Schöneberg, M. Simonović, J. Lesgourgues, and M. Zaldarriaga, Beyond the traditional line-of-sight approach of cosmological angular statistics, *Journal of Cosmology and Astroparticle Physics* **2018** (10), 047, [arXiv:1807.09540](#).
- [24] X. Fang, E. Krause, T. Eifler, and N. MacCrann, Beyond limber: efficient computation of angular power spectra for galaxy clustering and weak lensing, *Journal of Cosmology and Astroparticle Physics* **2020** (05), 010, [arXiv:1911.11947 \[astro-ph.CO\]](#).
- [25] S. Pandey, E. Krause, B. Jain, N. MacCrann, J. Blazek, M. Crocce, J. DeRose, X. Fang, I. Ferrero, O. Friedrich, *et al.*, Perturbation theory for modeling galaxy bias: Validation with simulations of the Dark Energy Survey, *Physical Review D* **102**, 123522 (2020), [arXiv:2008.05991 \[astro-ph.CO\]](#).
- [26] E. Krause, X. Fang, S. Pandey, L. F. Secco, *et al.*, Dark Energy Survey Year 3 Results: Multi-Probe Modeling Strategy and Validation, arXiv e-prints, [arXiv:2105.13548 \(2021\)](#), [arXiv:2105.13548 \[astro-ph.CO\]](#).
- [27] X. Fang, T. Eifler, and E. Krause, 2d-fftlog: efficient computation of real-space covariance matrices for galaxy clustering and weak lensing, *Monthly Notices of the Royal Astronomical Society* **497**, 2699 (2020), [arXiv:2004.04833 \[astro-ph.CO\]](#).
- [28] Z. Slepian, Y. Li, M. Schmittfull, and Z. Vlah, Rotation method for accelerating multiple-spherical bessel function integrals against a numerical source function, arXiv e-prints (2019), [arXiv:1912.00065 \[astro-ph.CO\]](#).
- [29] K. Meigs and Z. Slepian, On a general method for resolving integrals of multiple spherical bessel functions against power laws into distributions, arXiv e-prints (2021),

- arXiv:2112.07809 [math.CA].
- [30] J. Chellino and Z. Slepian, Triple-spherical bessel function integrals with exponential and gaussian damping: Towards an analytic N -point correlation function covariance model, *Proceedings of the Royal Society A* **479**, 20230138 (2023), arXiv:2308.01955 [math-ph].
 - [31] DLMF, NIST Digital Library of Mathematical Functions, <https://dlmf.nist.gov/>, Release 1.2.8 of 2026-09-15 (2026), f. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds.
 - [32] A. Erdélyi, W. Magnus, F. Oberhettinger, and F. G. Tricomi, *Tables of Integral Transforms. Vol. I* (McGraw-Hill, New York, 1954).
 - [33] A. Bzowski, P. McFadden, and K. Skenderis, Implications of conformal invariance in momentum space, *Journal of High Energy Physics* **2014**, 111 (2014), arXiv:1304.7760 [hep-th].
 - [34] A. Bzowski, P. McFadden, and K. Skenderis, Scalar 3-point functions in CFT: renormalisation, beta functions and anomalies, *Journal of High Energy Physics* **2016**, 066 (2016), arXiv:1510.08442 [hep-th].
 - [35] S. Albayrak and S. Kharel, Cosmological Correlators from Scattering Amplitudes: A Review, arXiv e-prints, arXiv:2609.19252 (2026), arXiv:2609.19252 [hep-th].
 - [36] Planck Collaboration, Planck 2018 results. VI. Cosmological parameters, *Astronomy and Astrophysics* **641**, A6 (2020), arXiv:1807.06209 [astro-ph.CO].
 - [37] D. J. Eisenstein and W. Hu, Baryonic Features in the Matter Transfer Function, *The Astrophysical Journal* **496**, 605 (1998), arXiv:astro-ph/9709112 [astro-ph].
 - [38] J. N. Fry and E. Gaztanaga, Biasing and hierarchical statistics in large-scale structure, *The Astrophysical Journal* **413**, 447 (1993).
 - [39] Z. Slepian, D. J. Eisenstein, F. Beutler, C.-H. Chuang, A. J. Cuesta, J. Ge, H. Gil-Marín, S. Ho, F.-S. Kitaura, C. K. McBride, R. C. Nichol, W. J. Percival, S. Rodríguez-Torres, A. J. Ross, R. Scoccimarro, H.-J. Seo, J. Tinker, R. Tojeiro, and M. Vargas-Magaña, The large-scale three-point correlation function of the SDSS BOSS DR12 CMASS galaxies, *Monthly Notices of the Royal Astronomical Society* **468**, 1070 (2017), arXiv:1512.02231 [astro-ph.CO].
 - [40] N. S. Sugiyama, S. Saito, F. Beutler, and H.-J. Seo, A complete FFT-based decomposition formalism for the redshift-space bispectrum, *Monthly Notices of the Royal Astronomical Society* **484**, 364 (2019), arXiv:1803.02132 [astro-ph.CO].
 - [41] N. S. Sugiyama, D. Yamauchi, T. Kobayashi, T. Fujita, S. Arai, S. Hirano, S. Saito, F. Beutler, and H.-J. Seo, New constraints on cosmological modified gravity theories from anisotropic three-point correlation functions of BOSS DR12 galaxies, *Monthly Notices of the Royal Astronomical Society* **523**, 3133 (2023), arXiv:2302.06808 [astro-ph.CO].
 - [42] Euclid Collaboration, Euclid preparation. LXXVIII. Full-shape modelling of two-point and three-point correlation functions in real space, *Astronomy and Astrophysics* **707**, A228 (2026), arXiv:2506.22257 [astro-ph.CO].
 - [43] O. Umeh, Optimal computation of anisotropic galaxy three point correlation function multipoles using 2DFFT-LOG formalism, *Journal of Cosmology and Astroparticle Physics* **2021** (05), 035, arXiv:2011.05889 [astro-ph.CO].
 - [44] M. Guidi, A. Veropalumbo, E. Branchini, A. Eggeimer, and C. Carbone, Modelling the next-to-leading order matter three-point correlation function using FFTLog, *Journal of Cosmology and Astroparticle Physics* **2023** (08), 066, arXiv:2212.07382 [astro-ph.CO].
 - [45] K. Pardede, F. Rizzo, M. Biagetti, E. Castorina, E. Sefusatti, and P. Monaco, Bispectrum-window convolution via Hankel transform, *Journal of Cosmology and Astroparticle Physics* **2022** (10), 066, arXiv:2203.04174 [astro-ph.CO].
 - [46] M. S. Wang, F. Beutler, J. Aguilar, S. Ahlen, D. Bianchi, D. Brooks, T. Claybaugh, A. de la Macorra, P. Doel, A. Font-Ribera, E. Gaztañaga, G. Gutierrez, K. Honscheid, C. Howlett, D. Kirkby, A. Lambert, M. Landriau, R. Miquel, G. Niz, F. Prada, I. Pérez-Ràfols, G. Rossi, E. Sanchez, D. Schlegel, M. Schubnell, D. Sprayberry, G. Tarlé, and B. A. Weaver, Window convolution of the galaxy clustering bispectrum, *Journal of Cosmology and Astroparticle Physics* **2025** (06), 031, arXiv:2411.14947 [astro-ph.CO].
 - [47] L. Fu, M. Kilbinger, T. Erben, C. Heymans, H. Hildebrandt, H. Hoekstra, T. D. Kitching, Y. Mellier, L. Miller, E. Semboloni, P. Simon, L. Van Waerbeke, J. Coupon, J. Harnois-Déraps, M. J. Hudson, K. Kuijken, B. Rowe, T. Schrabback, S. Vafaei, and M. Velander, CFHTLenS: Cosmological constraints from a combination of cosmic shear two-point and three-point correlations, *Monthly Notices of the Royal Astronomical Society* **441**, 2725 (2014), arXiv:1404.5469 [astro-ph.CO].
 - [48] L. F. Secco, M. Jarvis, B. Jain, *et al.*, Dark Energy Survey Year 3 Results: Three-Point Shear Correlations and Mass Aperture Moments, *Physical Review D* **105**, 103537 (2022), arXiv:2201.05227 [astro-ph.CO].
 - [49] R. C. H. Gomes, S. Sugiyama, B. Jain, M. Jarvis, *et al.*, Dark Energy Survey Year 3 Results: Cosmological constraints from second- and third-order shear statistics, *Physical Review D* **112**, 123515 (2025), arXiv:2508.14018 [astro-ph.CO].
 - [50] S. Sugiyama, R. C. H. Gomes, and M. Jarvis, Fast modeling of the shear three-point correlation function, *Physical Review D* **113**, 123548 (2026), arXiv:2407.01798 [astro-ph.CO].
 - [51] O. Friedrich, F. Andrade-Oliveira, H. Camacho, *et al.*, Dark Energy Survey year 3 results: covariance modelling and its impact on parameter estimation and quality of fit, *Monthly Notices of the Royal Astronomical Society* **508**, 3125 (2021), arXiv:2012.08568 [astro-ph.CO].
 - [52] DES Collaboration, T. M. C. Abbott, *et al.*, Dark Energy Survey Year 3 Results: Cosmological Constraints from Galaxy Clustering and Weak Lensing, *Physical Review D* **105**, 023520 (2022), arXiv:2105.13549 [astro-ph.CO].
 - [53] O. H. E. Philcox, Probing Parity-Violation with the Four-Point Correlation Function of BOSS Galaxies, *Physical Review D* **106**, 063501 (2022), arXiv:2206.04227 [astro-ph.CO].
 - [54] J. Hou, Z. Slepian, and R. N. Cahn, Measurement of parity-odd modes in the large-scale 4-point correlation function of SDSS BOSS DR12 CMASS and LOWZ galaxies, *Monthly Notices of the Royal Astronomical Society* **522**, 5701 (2023), arXiv:2206.03625 [astro-ph.CO].
 - [55] A. Krolewski, S. May, K. Smith, and H. Hopkins, No evidence for parity violation in BOSS, *Journal of Cosmology and Astroparticle Physics* **2024** (08), 044, arXiv:2407.03397 [astro-ph.CO].
 - [56] M. Reinhard, Z. Slepian, J. Hou, and A. Greco, Full Parity-

- Violating Trispectrum in Axion Inflation: Reduction to Low-D Integrals, arXiv e-prints , arXiv:2412.16037 (2024), [arXiv:2412.16037 \[astro-ph.CO\]](#).
- [57] J. Hou, R. N. Cahn, O. H. E. Philcox, and Z. Slepian, Analytic gaussian covariance matrices for galaxy N -point correlation functions, *Physical Review D* **106**, 043515 (2022), [arXiv:2108.01714 \[astro-ph.CO\]](#).
 - [58] W. J. Percival, D. Burkey, A. Heavens, A. Taylor, S. Cole, J. A. Peacock, *et al.*, The 2dF Galaxy Redshift Survey: Spherical Harmonics analysis of fluctuations in the final catalogue, *Monthly Notices of the Royal Astronomical Society* **353**, 1201 (2004), [arXiv:astro-ph/0406513 \[astro-ph\]](#).
 - [59] H. S. Grasshorn Gebhardt and O. Doré, Fabulous code for spherical Fourier-Bessel decomposition, *Physical Review D* **104**, 123548 (2021), [arXiv:2102.10079 \[astro-ph.CO\]](#).
 - [60] B. Khek, H. S. Grasshorn Gebhardt, and O. Doré, Fast Theoretical Predictions for Spherical Fourier Analysis of Large-Scale Structures, *Physical Review D* **110**, 063524 (2024), [arXiv:2212.05760 \[astro-ph.CO\]](#).
 - [61] E. Castorina and M. White, Beyond the plane-parallel approximation for redshift surveys, *Monthly Notices of the Royal Astronomical Society* **476**, 4403 (2018), [arXiv:1709.09730 \[astro-ph.CO\]](#).
 - [62] E. Castorina and M. White, The Zeldovich approximation and wide-angle redshift-space distortions, *Monthly Notices of the Royal Astronomical Society* **479**, 741 (2018), [arXiv:1803.08185 \[astro-ph.CO\]](#).
 - [63] E. Castorina and M. White, Wide-angle effects for peculiar velocities, *Monthly Notices of the Royal Astronomical Society* **499**, 893 (2020), [arXiv:1911.08353 \[astro-ph.CO\]](#).
 - [64] J. N. Benabou, A. Testa, C. Heinrich, H. S. Grasshorn Gebhardt, and O. Doré, Galaxy bispectrum in the spherical fourier–bessel basis, *Physical Review D* **109**, 103507 (2024), [arXiv:2312.15992 \[astro-ph.CO\]](#).
 - [65] DESI Collaboration, DESI 2024 VII: Cosmological Constraints from the Full-Shape Modeling of Clustering Measurements, *Journal of Cosmology and Astroparticle Physics* **2025** (07), 028, [arXiv:2411.12022 \[astro-ph.CO\]](#).
 - [66] J. J. M. Carrasco, S. Foreman, D. Green, and L. Senatore, The Effective Field Theory of Large Scale Structures at Two Loops, *Journal of Cosmology and Astroparticle Physics* **2014** (07), 057, [arXiv:1310.0464 \[hep-th\]](#).
 - [67] M. Fasiello, T. Fujita, and Z. Vlah, Perturbation theory of large scale structure in the Λ CDM Universe: Exact time evolution and the two-loop power spectrum, *Physical Review D* **106**, 123504 (2022), [arXiv:2205.10026 \[astro-ph.CO\]](#).
 - [68] Z. Slepian, On decoupling the integrals of cosmological perturbation theory, *Monthly Notices of the Royal Astronomical Society* **507**, 1337 (2021), [arXiv:1812.02728 \[astro-ph.CO\]](#).